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F. Greg Shinskey on External Reset

Greg McMillan on Advanced Process Control

The Dow Automation Learnings, Part Two



Unscrambling SAFETY

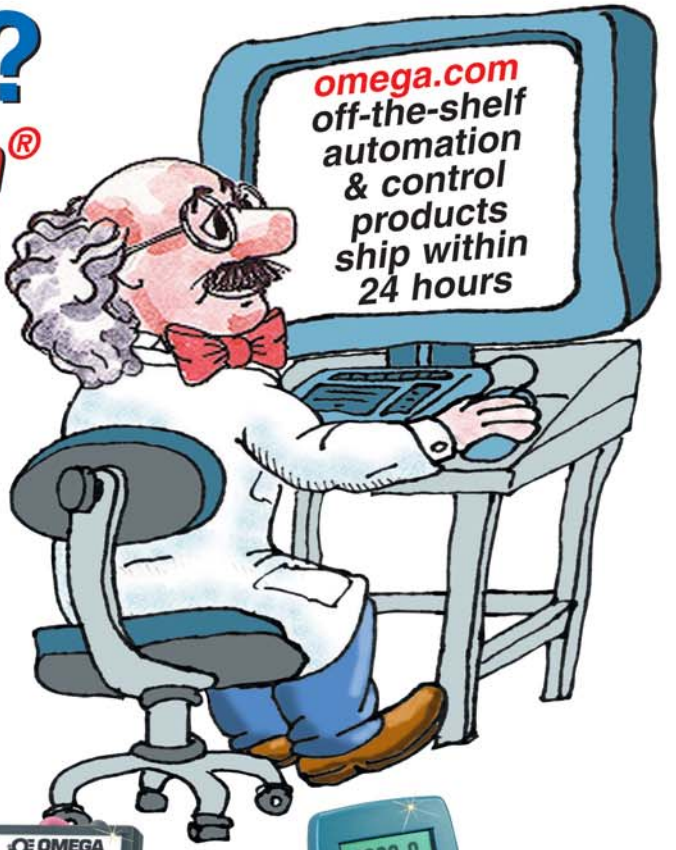
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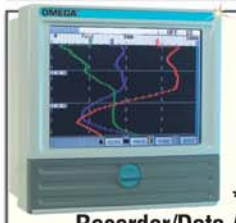
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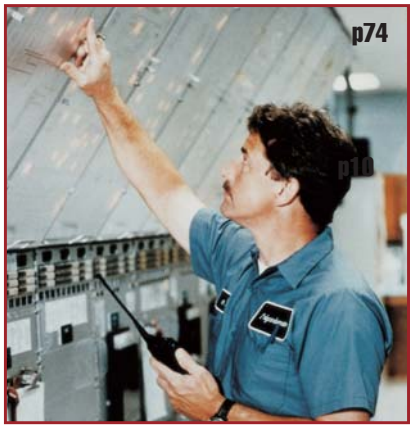
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Dow Chemical's transition to a new development approach, with some help from ABB.

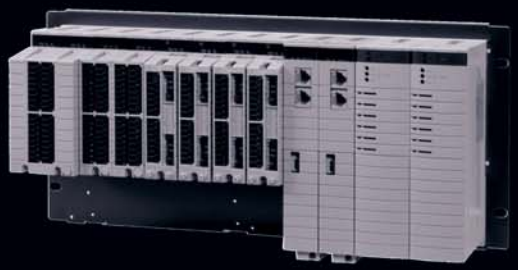
Margaret Walker, Ed Sederlund, Jerry Gipson, and Eric Cosman, Dow Chemical Co.

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CIRCULATION AUDITED JUNE/2005

Chemicals & Allied Products	15,926
Food & Kindred Products	15,821
Paper & Allied Products	4,087
Primary Metal Industries	5,341
Electric, Gas & Sanitary Services	2,707
System Integrators & Engineering Design Firms	7,933
Rubber and Miscellaneous Plastic Products	5,016
Stone, Clay, Glass & Concrete products	2,580
Textile Mill Products	2,088
Petroleum Refining & Related Industries	3,361
Tobacco Products	167
Total circulation	65,027

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Reason #68 of over 200:

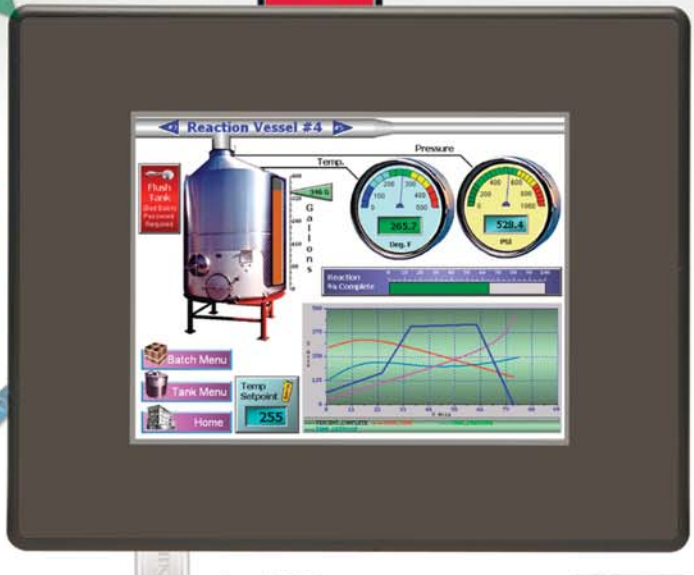
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Immigrants Are Us

I'm talking politics here. Process automation isn't immune, and there's something going on that is very relevant to what we'll do in the future, as well as what we've done in the past.

Congress and the nation are mired in a debate over the status of undocumented immigrants to the U.S. I doubt sincerely that by the time you read this, the debate will be over, or a solution found. That's too bad.



The globalization of the world economy is already done. National interests are inconvenient holdovers from the last century, and corporations will go where the money is, where the engineering talent is, and where the business takes them.

However, the workforce isn't global. It's only a matter of time before the tsunami of parity rolls all the way around the world. Workers in India are now getting close to parity (adjusted for cost of living) with workers in the First World. This will happen everywhere. It's much easier for a company to be global than it is for a worker to pull up stakes, and seek better opportunities elsewhere. And yet they do, and have done so for hundreds of years.

My maternal grandparents came to the U.S. from Italy about 1910. At that time, the requirements to come to the U.S. were so much less stringent that a man and woman from Italy could just get on a boat and come to America, looking for a better life. Both of them spoke Italian all their lives, preferentially. My grandfather, Stefano Martini, raised his family, lived and died in the U.S. I don't know if he ever became a citizen. My grandmother, Marietta Martini, did become a citizen. In 1910, in the Santa Clara Valley, Italians were the "wops" who picked the crops, worked in the fields, worked in the canneries, and cleaned the houses. Now, in Silicon Valley, those jobs are held by Latin Americans, mostly undocumented. In the 19th Century, we discriminated against immigrants, because they "would change our way of life." Well, the pot melted, and they (we) are all Americans now. This will happen again. Even the "Native Americans" aren't really. They came over the land bridge from Siberia earlier than my grandparents came through Ellis Island. We are all immigrants here.

My other grandfather, Joseph Denim Boyes, was born in Glasgow, Scotland. He immigrated to Canada with his family in the 1890s, and came down to the U.S. to work on the railroad in Idaho. Eventually, he became a U.S. citizen. He married a girl, Grace Barr, whose family was Pennsylvania Dutch, immigrants from Germany who crossed the country in a covered wagon looking for opportunity. Between them, they raised eight children, including my father, who went to work for Brown Instruments in 1940, and bequeathed to me the world of automation.

None of my grandparents were ever told they couldn't come here. We were the land of opportunity, then. We should still be the land of opportunity now. The amount of innovation produced by immigrants is staggering. Without those immigrants, there simply would not be a United States as we know it.

What does this mean for process automation? Where are we going to get those younger workers that we need, without whom jobs are already going begging?

That's right. You guessed it.



***Walt Boyes,** editor in chief*
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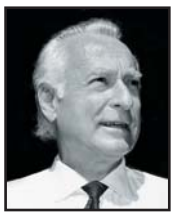
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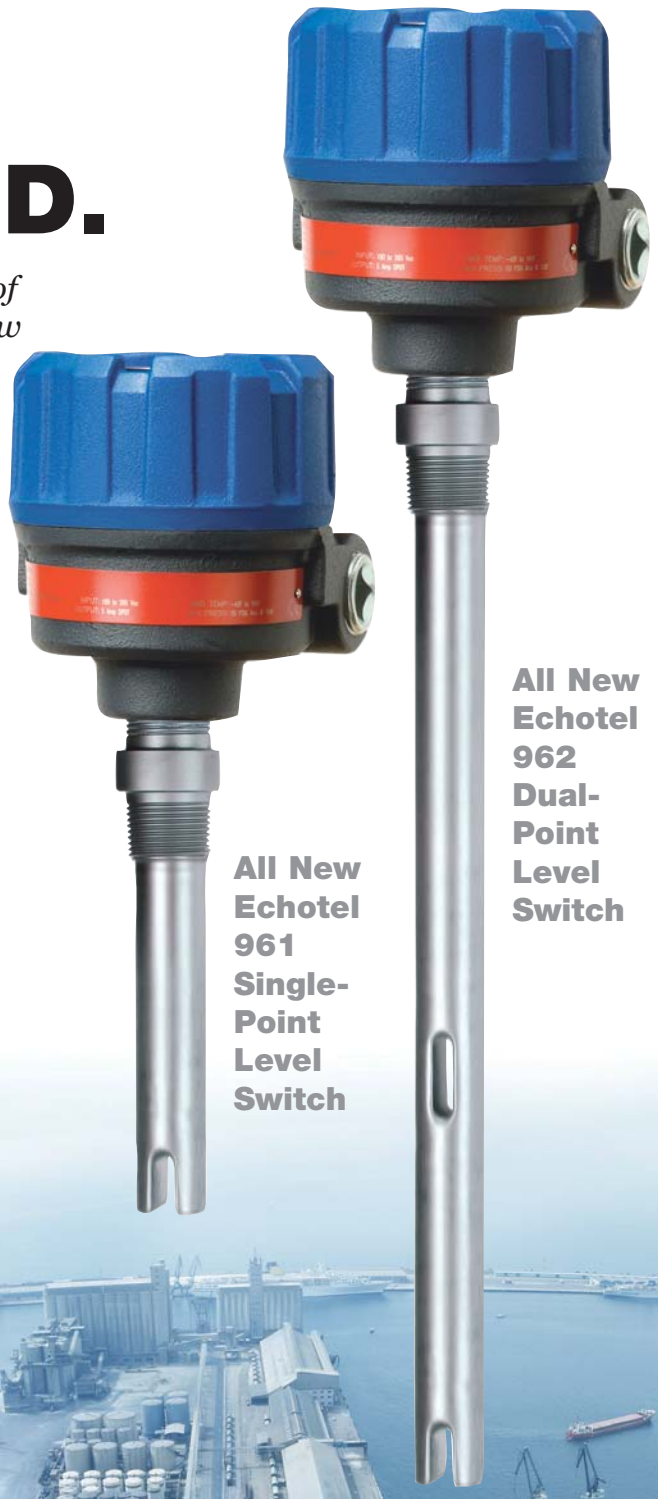
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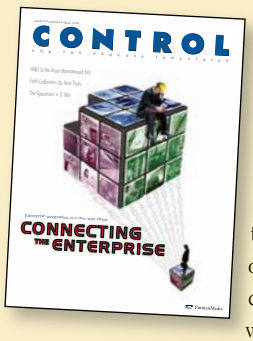
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Replace Junk Arguments With Global Warming Facts

In 1633, when Galileo's discoveries were called "junk science," he reportedfly replied: "E pur si muove! (And yet, it moves!)



The goal of my three-part article (see "A Process Only Mankind Can Control, Part 3" on p. 16) is *not* to convince anyone that mankind is causing global warming. My purpose is to show that the knowledge that we in the process control profession have accumulated during the last century can be very useful in understanding this heat storage/transfer (or any other) process.

I've shown that the Earth's heat balance is neither inherently stable nor self-regulating, but is a large-inertia, integrating process. I also described such positive feedbacks as the melting of the icecaps' permafrost, which changes the role of arctic regions from cooling the globe (solar reflector) to a heat source (solar absorber).

Process control people know that non-self-regulating processes, if left uncontrolled, will "run away." We also know that, presently, this process operates in open loop, not in "manual!" In manual control, the loop has a manually implemented setpoint. This Earth's loop does not! Its loop unfortunately is open because some people (like my fellow columnist Rich Merritt, in his response the "Global Warming Merits Extra Caution" letter in *Control*, March '06, p. 15) believe this process is self regulating, and that it will control itself. Similarly, there is no global agreement yet on what variables should or can be manipulated to bring the global temperature control loop back to setpoint.

The beauty of process control's artificial neural networks (ANN) is their ability to quantitatively determine the gains, time constants, and nonlinearities of this complex, multivariable process. ANN models can then be trained on on historical data, so they can help predict the future. Such predictions can involve, not only such simple relationships as the rise of the sea levels as a function of the melting of the arctic ice cover (if all polar ice melted, ocean levels would rise more than 200 feet), but also the more complex relationships, such as the affect of global population, GDP, car mileage, etc. on global warming. In short, a trained process control model can quantify the consequences of both inaction and corrective action, and can, thereby, replace "junk arguments" with scientific facts.

Béla G. Lipták, PE
Control columnist

Finer Valve Resolution

I recently read your article, "A Fine Time to Break Away from Old Valve Problems," *Control*, Nov. '05, p. 57, and found it very interesting. I agree with Greg McMillan saying that fine adjustment of the valve would control the valve resolution. MEA, has been building Electro-Hydraulic actuators for 40 years, and we've solved the problem of requiring two valves for control rangeability many times. Once, we replaced two CCI valves with pneumatic actuators in a GE-based nuclear water level control with MEA's actuators. Though they used to start with the small valve and switch to the large one, the operators found they could start up with only the large valve, and still have excellent control. The level control also was improved with 0.1% resolution, 0.1% linearity, 5 Hz frequency response. and 0.1% positionng accuracy. The combination of frequency response accuracy provides the best valve resolution.

T.W. Lasiewicz, president
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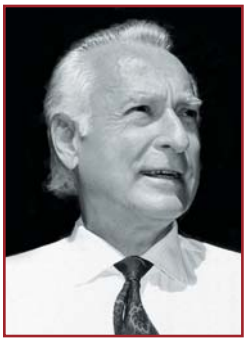
A Process Only Mankind Can Control, Part 3

In the past, life on Earth was sustained by solar energy alone (Figure 1). This inexhaustible and clean energy was used by the world's green plants, which converted carbon dioxide into organic material, while releasing oxygen. Animal life completed this cycle, obtaining muscle energy by oxidizing the plants, while inhaling oxygen and exhaling carbon dioxide. Therefore, so long as the global density of plant and animal life was balanced, the carbon dioxide content of the atmosphere remained constant.

In the present, modern man needs more energy for comfort, transportation, etc. However, energy obtained from green plants is no longer sufficient, and is supplemented by exhaustible hydrocarbon fossil deposits. Naturally, burning the planet's fossil energy deposits increases carbon dioxide and other gas concentrations in the atmosphere. These greenhouse gases block reflection of thermal radiation back into outer space. The result is global warming. It's like placing a blanket over the planet.

The Solution

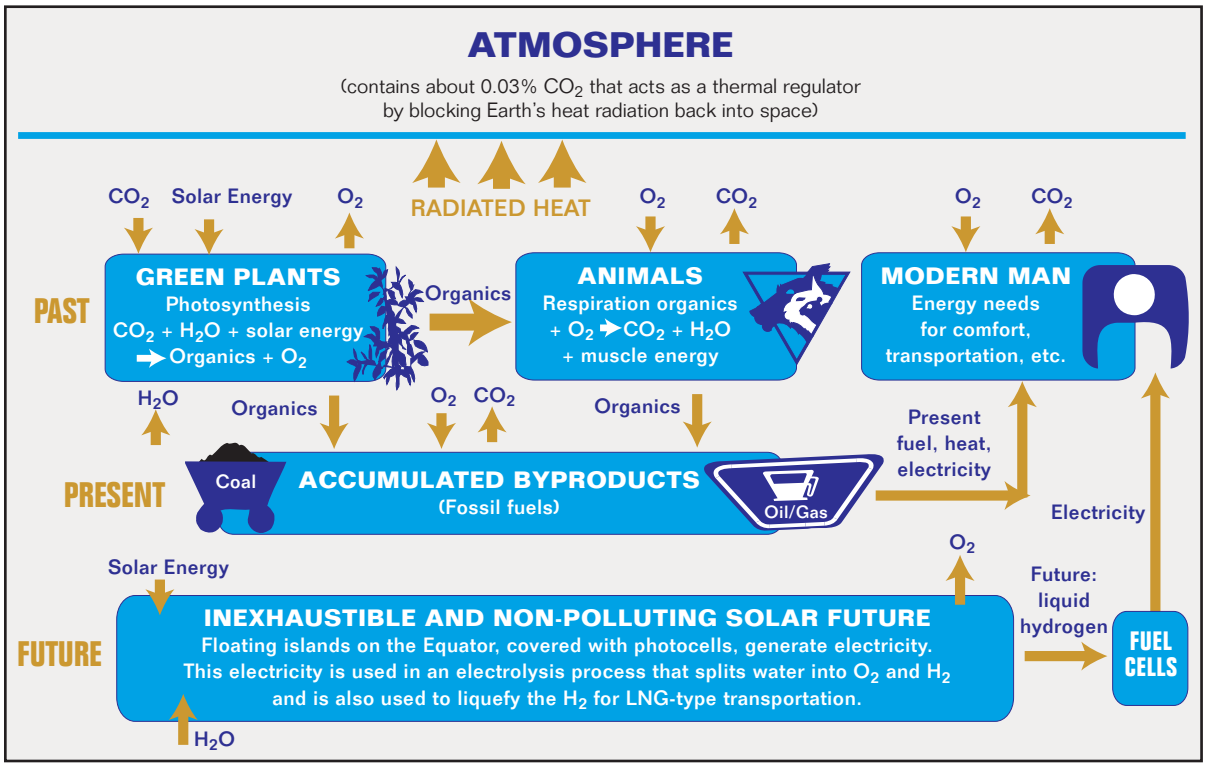
There is little question that the only inexhaustible energy source is the Sun. Solar energy can be collected at the equator—on land in the Sahara or on floating islands in the ocean—and can be converted into electrical energy by photocells. This electricity can be stored in the form of chemical energy by splitting the water into oxygen and hydrogen (electrolysis). The oxygen can than be released into the atmosphere (just as plants do), while the liquefied hydrogen can be distributed, just as LNG is distributed today.



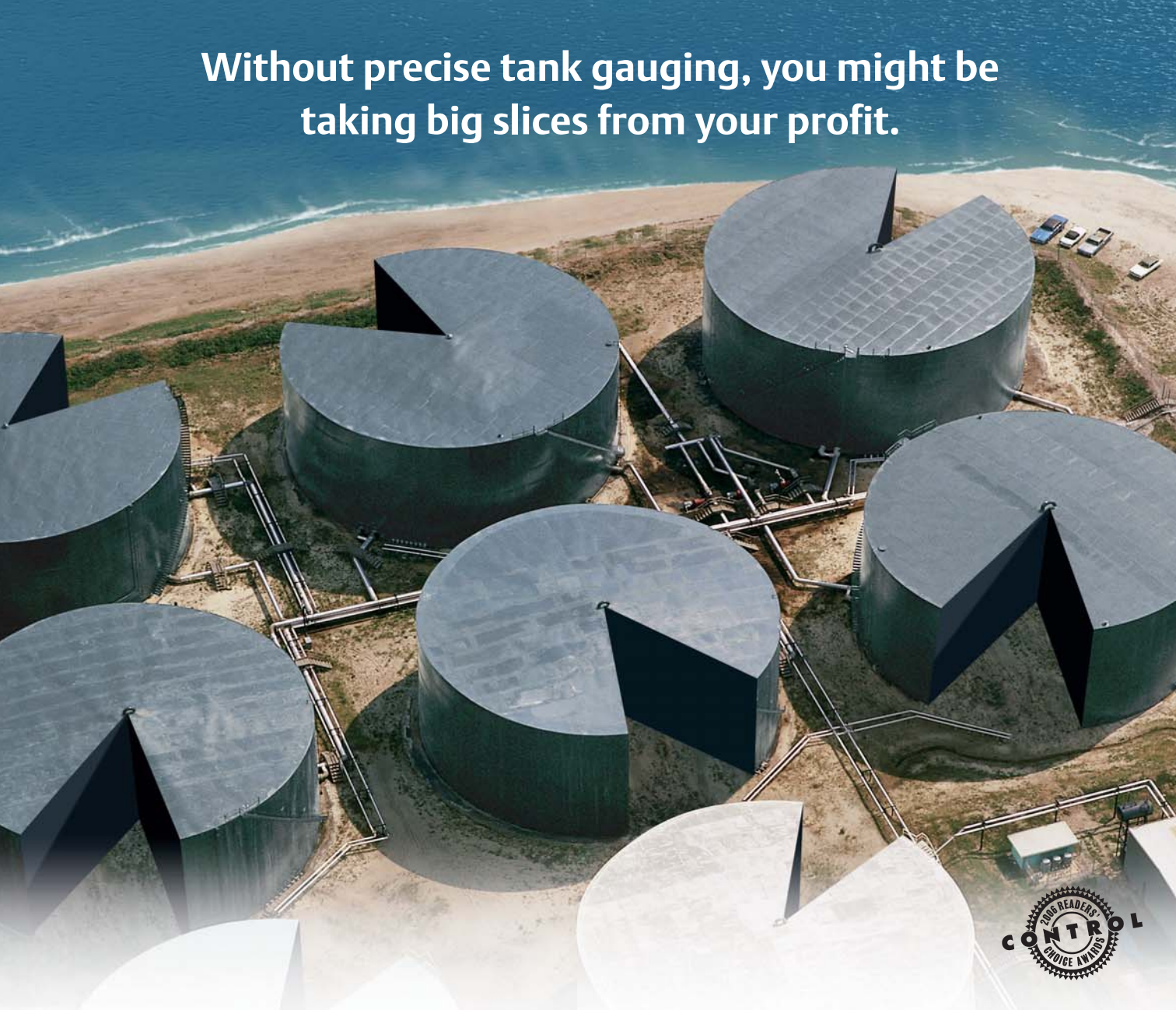
It's consistent with our irresponsible age to overestimate the time available to convert to a clean and inexhaustible energy.

FIGURE 1.

CARBON DIOXIDE'S ROLE IN GLOBAL WARMING



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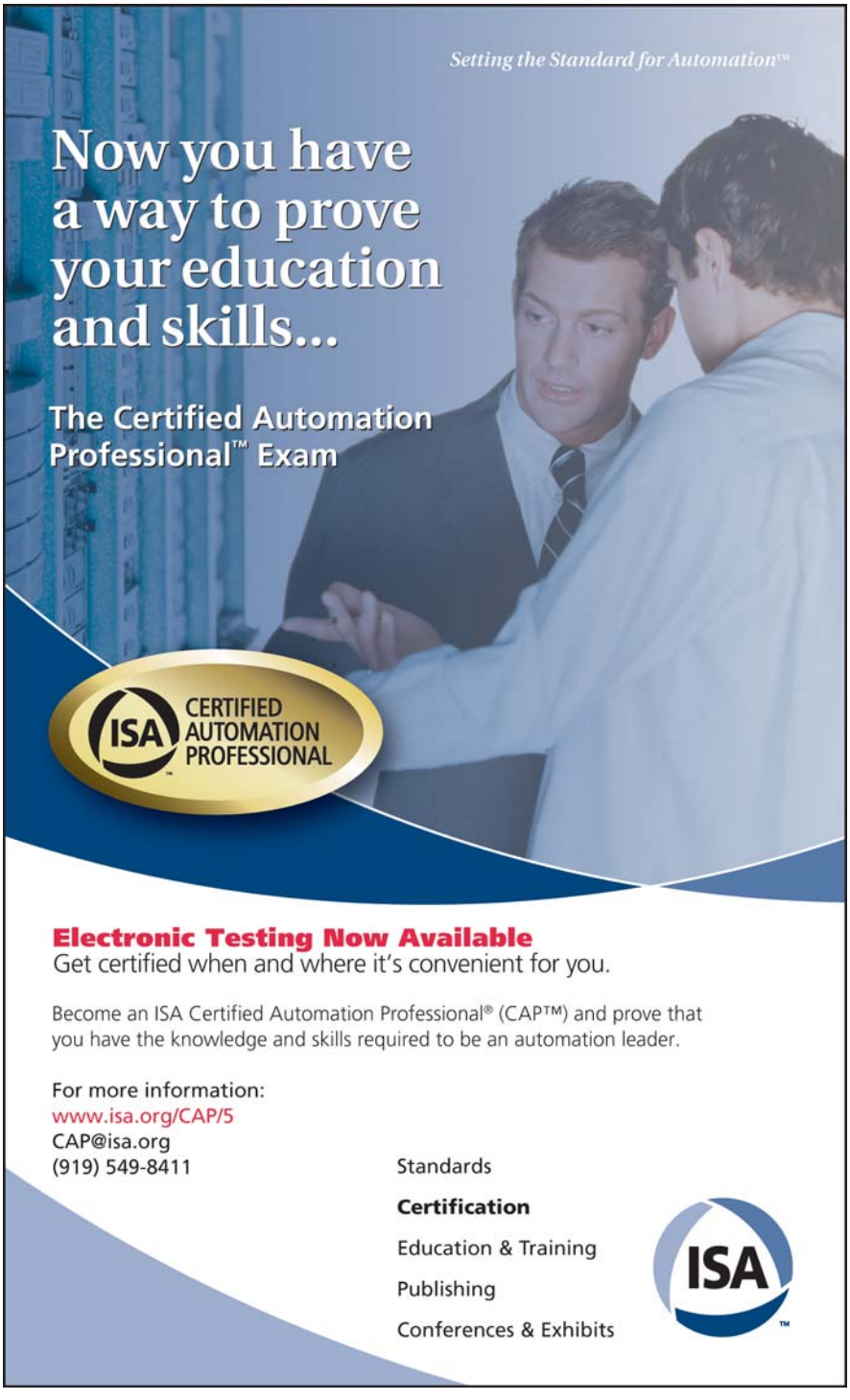
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When the hydrogen is received by users, fuel cells can reconvert its chemical energy back into electricity. In the future, hydrogen can fuel our homes, power plants, and transport vehicles, just as it fuels spacecraft today. This "hydrogen economy" would not only be clean and inexhaustible, but would also eliminate global warming by stopping the generation of carbon dioxide.

An indirect way of using solar energy is to produce hydrogen from agricultural products, such as glucose or ethanol. This approach is inferior to using water as the source of hydrogen because generating hydrogen from organic materials again produces greenhouse gases. In Figure 1, solar energy is stored in hydrogen's chemical energy, but only water is produced when it's reconverted to release electric energy.



The Time Factor

Some argue that we have plenty of time because fossil fuels will last for generations, and because the causes of global warming are debatable. Those who share this viewpoint are likely to promote more oil exploration, and building more nuclear and coal-burning power plants. They're likely to propose waiting until we either run out of all these resources, or until the consequences of nuclear waste accumulation and/or global warming become intolerable.

We live in an irresponsible age: the population of the planet has increased 500% during the last century, the stockpile of nuclear warheads is still staggering, and our government's budget is balanced at the expense of coming generations. It's consistent with this attitude of irresponsibility to overestimate the time available to convert to a clean and inexhaustible energy source. Unfortunately, as was shown in the discussion of the processes that guide the behavior of ocean currents, the time available to make this decision, can be much shorter than we think.

[Editor's note: this is the third installment of Béla Lipták's three-part series on using process control principles to understand and perhaps control global warming. The first part, "A Process Only Mankind Can Control, Part 1," appeared in Control, *Jan. '06, p. 20. The second part, "Can Process Control Stabilize Global Warming?" appeared in* Control, *March '05, p. 16.]*

Béla Lipták, PE, process control consultant, is also editor of the Instrument Engineers' Handbook. He can be reached at liptakbela@aol.com.

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The Biggest Hurdle?

To continue our coverage of challenges faced by new technologies, particularly automation technologies and fieldbuses, another hurdle to their adoption is skid and modular-equipment manufacturers. This is a double whammy because most new facilities are no longer "stick built," but are entirely modular, so end users often can't benefit from digital field devices. The second half of the whammy is that fieldbuses are well suited to modular design and fabrication because they can be preconfigured only as required. Fieldbus technologies lend themselves to slight modifications with no impact to the inter-module connections because the instruments are all wired to the same home-run, twisted-pair cabling. Taking this potential expansion into consideration on the base design maximizes flexibility for the price of a short cable run and the instrument itself.

In addition, because all manufacturers must remain competitive, the instrumentation selected is often the simplest, lowest cost, or *de facto* industry standard device that can provide the required measurement signal. With multivariable fieldbus transmitters, the number of expensive field devices can actually be reduced, thus saving not only the price of the instrument but also the nozzle and related connections too.



Many skid fabricators are less likely to change from traditional wiring to Foundation fieldbus due to unfamiliarity, increased cost, and perceived risk.

Unfortunately, not many skid or module manufacturers have their own in-house automation team, and even fewer have staffs that understand how fieldbus systems are designed. This is because mechanical and process engineering are the bread-and-butter, core part of the business, while electrical and Instrumentation (E&I)) are simply needed to complete the package. Though there aren't too many outside consultants available either, perhaps they could help fill this gap?

As indicated above, because fieldbus systems use a trunk-and-spur or star-type configuration, the number of connections/cables that need to cross the module or skid boundaries, which are the trunks or home-run cables, are connected to the base of these branches to complete the network. If the manufacturer is presently offering some form of remote I/O, then they're trying to take advantage of minimizing the home-run cable. However, as has been proven by the FURIOS study conducted by Adventis in Germany, fieldbus technologies are more economical than the remote I/O option.

Building skids with minimal field terminations also improves project quality because more assembly can be completed in a controlled shop environment, which also reduces installation costs. For skid or modular assembly manufacturer, who want to differentiate themselves from the crowd, while making larger profits, offering digital communications and fieldbus technology could be the ticket.

Ian Verhappen, contributing editor
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The Power of Ethernet Powerlink

By John Mazurkiewicz
Baldor Electric Company

The use of high-speed serial communication networks in factories is becoming commonplace, making it easy to integrate electronic motion control into the automation process. One generic high-speed communication platform receiving significant attention is Ethernet.



Standard Ethernet, while powerful, requires techniques for circumventing data collision on the line. IEEE 802.3 terms this "Carrier Sense Multiple Access Collision Detection" (CSMA/CD). Each Ethernet device connects to the network with a single cable through an infrastructure component (such as an Ethernet switch). This then sends the communication to the particular Ethernet device, which has a specific "Media Access Control" or MAC address. All addresses are globally unique because each manufacturer of Ethernet devices is assigned a block of MAC IDs. Combined with IP addresses, the destination address of a device is known.

ETHERNET Powerlink

The success of Powerlink is due to its openness. Since it does not require custom chipsets or ASICs, the protocol uses standard chips that can be purchased from a wide choice of suppliers. At this time, it is supported by more than 300 leading industrial companies worldwide. ETHERNET Powerlink was developed from the field device up. This allows for seamless integration in the transfer of data from higher order IT systems, down to drives and sensors, even Internet access to motion control on the shop floor, for remote control and diagnostics. Ethernet allows thousands of devices to be interconnected on the same network, compared to others that allow a limited range of connected devices. These features eliminate the need for different multiple network technologies.

Powerful Automation Packages

Modern industrial automation motion control packages are powerful, flexible, easy to use, and offer a host of capabilities. Designing around a network interface like ETHERNET Powerlink, fewer centralized motion controllers will be used, thus reducing cost. In addition, Powerlink DS402 decentralized drives can perform velocity or torque and also positioning tasks, such as absolute and relative moves, homing, etc. In terms of software benefits, motion controllers using comprehensive basic type languages provide ease of use, reduce development time, helping to get the product to market faster. Baldor's popular multi-tasking MINT is available for use with Windows hosts, or 'C' in embedded form, and ActiveX. Multi-tasking capability allows for control capability of motion, I/O, logic control, HMI and communication tasks. This single programming tool also allows complex applications to be broken down into simpler, more manageable sub-tasks, making the overall job easier.

Proven Results

In applications such as packaging, fabrication, cut-to length, etc., the machines often involve numerous servo axes. At the same time, the customer needs a solution to reduce overall production costs. In one such application, involving over 15 servo motors and six different tools, the challenge was to combine all into a single operating set up while automatically moving the required item into place without shut down or pausing other upstream stations feeding this machine. The solution involved utilizing ETHERNET Powerlink and decentralized motion control. Critical axes were updated every 250 microseconds. ETHERNET Powerlink easily handles this rate (which is significantly faster than cycle times of other communication systems). Secondary axes can be controlled at slower rates.



This method improved the machine throughput to over 200 percent up to 50 feet per minute, while improving tolerances to less than 0.04 inch. Baldor's brushless servo motors (BSM Series) provided fast, accurate positioning. By combining benefits of ETHERNET Powerlink and decentralized motion control, unprecedented process accuracy was attained despite large number of axes. The new machine produces higher quality parts and cut operating costs. Set-up time was reduced, scrap was reduced, and overall machine flexibility was increased.

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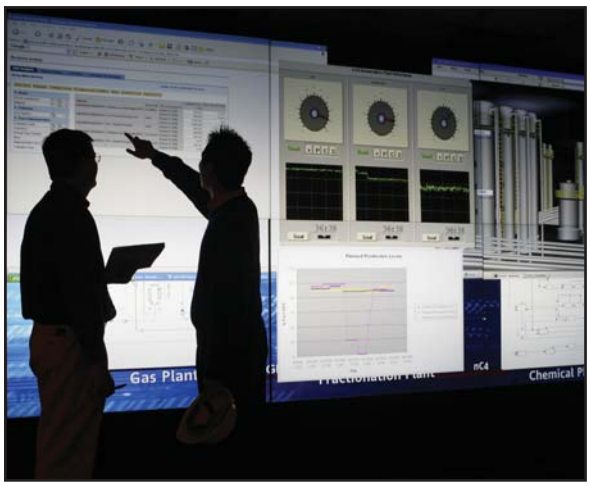
InFusion merges automation, information to enable unified control across plant, enterprise

Invensys has introduced what it reports is the first industrial system that goes beyond plant and industrial operations to provide a "true enterprise view." InFusion enterprise control system combines Invensys' capabilities with enterprise information and integration technologies from Microsoft and SAP to reduce integration costs. Most existing plant-floor and enterprise systems reportedly can use InFusion to cost-effectively integrate into a common system. The company adds that InFusion works with several new Invensys performance services to help enterprise users align plant operations and maintenance departments more effectively, and optimize overall asset management and performance.

"InFusion enterprise control system is as revolutionary as our Foxboro I/A Series automation system and Wonderware InTouch HMI software were when they were first introduced," says Mike Caliel, president of Invensys Process Systems. "By combining the latest capabilities from across Invensys into one, unified architecture, Invensys has realized a step change in using open technologies and standards in a fully industrial system. By dramatically reducing plant and enterprise integration costs, InFusion will help our customers finally break down stubborn technical and organizational barriers, while preserving and extending their equity in existing automation assets."

FIGURE 1.

COLLABORATION WALL



InFusion's Collaboration Wall HMI concept gives users a shared view of process control, maintenance performance, and business application displays.

With more than 60 issued and pending patents and patent applications, InFusion provides capabilities that Invensys says weren't previously available from one automation or information system, including:

- Integration across existing plant-floor (DCS, PLC, etc.) systems, subsystems, and intelligent field devices, regardless of vendor or protocol
- Low-cost, standards-based information interoperability between plant-floor, manufacturing execution systems (MESs), and enterprise systems
- Unified engineering and support environment across plant-floor and MES systems, and an application-object model that simplifies creation and reuse of equipment, unit, and plant-level strategies, while improving engineering productivity
- Real-time visibility into the plant and the business with a unified view across operations and maintenance domains to help align overall plant performance with business objectives

In essence, InFusion combines Foxboro's process control capabilities with Wonderware's HMI, plant intelligence, and device integration capabilities into an ArchestrA-based system. ArchestrA is the software architecture that Invensys built on .NET and other Microsoft technologies. InFusion also incorporates other Invensys and third-party capabilities, including:

- Triconex's triple modular redundant (TMR) safety and critical control
- Foxboro's remote terminal units (RTUs) and intelligent SCADA applications
- SimSci-Esscor's control, simulation, and process optimization
- Avantis' real-time condition monitoring and enterprise asset management
- Foxboro's intelligent pressure, level, flow, and electrochemical transmitters, valve positioners, and advanced, FDT-compliant diagnostic tools
- A comprehensive, managed approach to wireless technology that reportedly makes it practical and cost effective to incorporate wireless measurements into the system

Invensys reports that InFusion reduces the cost and effort required to integrate intelligent, plant-floor devices and systems via standards, such as OPC, and a portfolio of device drivers. Process control, MES, and even ERP applications can be combined into a common data model and HMI to deliver in-context information to plant personnel.

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Interoperability, Unified Development

Invensys adds it worked with technology partners, such as Microsoft and SAP, to reduce the time and effort previously required to integrate real-time plant systems with transactional enterprise systems. InFusion uses a platform-independent, web services-based enterprise architecture that approach. For instance, InFusion uses Microsoft's .NET and BizTalk Server 2004, SAP's NetWeaver and xMII, ISA's S95 standard for manufacturing-to-enterprise integration, MIMOSA's standard for maintenance-to-enterprise integration), and OPC (for real-time connectivity).

InFusion also represents the first major implementation of Open Operations & Maintenance (O&M), which is the industry-standard convergence of OPC, ISA S95, and MIMOSA. This approach eliminates the need to use conventional point-to-point solutions that are costly to implement and maintain, and inherently inflexible.

In addition, InFusion provides an engineering environment that reduces the time and effort required to configure and support intelligent devices and systems, and implement control and information strategies. InFusion's engineering environment includes a unified, system-wide integrated development environment (IDE) and object-based equipment, unit, and plant application object models. These "nested" application objects can be configured, instantiated, cloned, and re-used with minimum time and effort. Objects also can be bundled and reused in and across multiple plant sites, allowing for knowledge capture and consistent best practices.

Also, its Field Device Manager uses FDT and enhanced EDDL to allow device vendors to program graphic-rich applications that can be plugged into InFusion's environment. Maintenance staffs call up these applications when they want to analyze the health and performance of a field device model, run diagnostic tests, and archive test results.

For more information, visit www.InFusionECS.com.

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ISA Standard Manages Batch Data from Multiple Control Systems

ISA reports its new ISA-88.00.04-2006 standard, "Batch Control Part 4: Batch Production Records," is the latest in its series of batch control standards. The new standard provides a detailed definition for batch production records, establishing a reference model for developing applications for the storage and/or exchange of batch production records. Implementations based on the standard will allow retrieval, analysis, and reporting of selected batch production record data.

"This new Part 4 standard addresses the major problem of managing production information from multiple different control systems," says ISA-SP88's chairman, Dennis Brandl, of BR&L Consulting. "It defines a structure for preserving production information in a vendor-independent manner, and was designed to meet regulatory requirements related to data collection and retention. The planned XML implementation of the standard will give vendors and end users a valuable method for long-term retention of searchable production information."

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Though the new standard is intended primarily for batch processes, Brandl adds it also may be useful for other processes. The new standard is the fourth in ISA's Batch Control Standards series, which also includes:

- ANSI/ISA-88.01-1995, "Batch Control Part 1: Models and Terminology"
- ANSI/ISA88.00.02-2001, "Batch Control Part 2: Data Structures and Guidelines for Languages"
- ANSI/ISA88.00.03-2003, "Batch Control Part 3: General and Site Recipe Models and Representation"

For more information about ISA standards, visit www.isa.org/standards.

FF's Updated Interoperability Kit Allows Testing with 5.0 DD Files

The Fieldbus Foundation (FF, www.fieldbus.org) recently released its enhanced H1 Interoperability Test Kit (ITK) Version 5.0. The test kit has added functions enabling testing of Foundation fieldbus devices using Device Description (DD) 5.0 files.

FF's ITK 5.0 verifies the functionality of an H1 (31.25 kbps) device and its conformity with Foundation fieldbus' Function Block and Transducer Block specifications. As a tool for troubleshooting and debugging devices, the test kit also includes hardware and software required to ensure a manufacturer's complete device interoperability as specified by FF's official registration testing procedure. Developers can use H1 ITK 5.0 to run tests identical to those FF uses before submitting their device for official registration.

In addition, ITK 5.0 has been updated from previous versions with added functions, expanding its scope of interoperability testing to benefit Foundation fieldbus devices. The following test cases, including two new features, have been added and/or modified:

- Multi-bit alarm testing now tests for single and multi-bit alarms, which provides more thorough testing;

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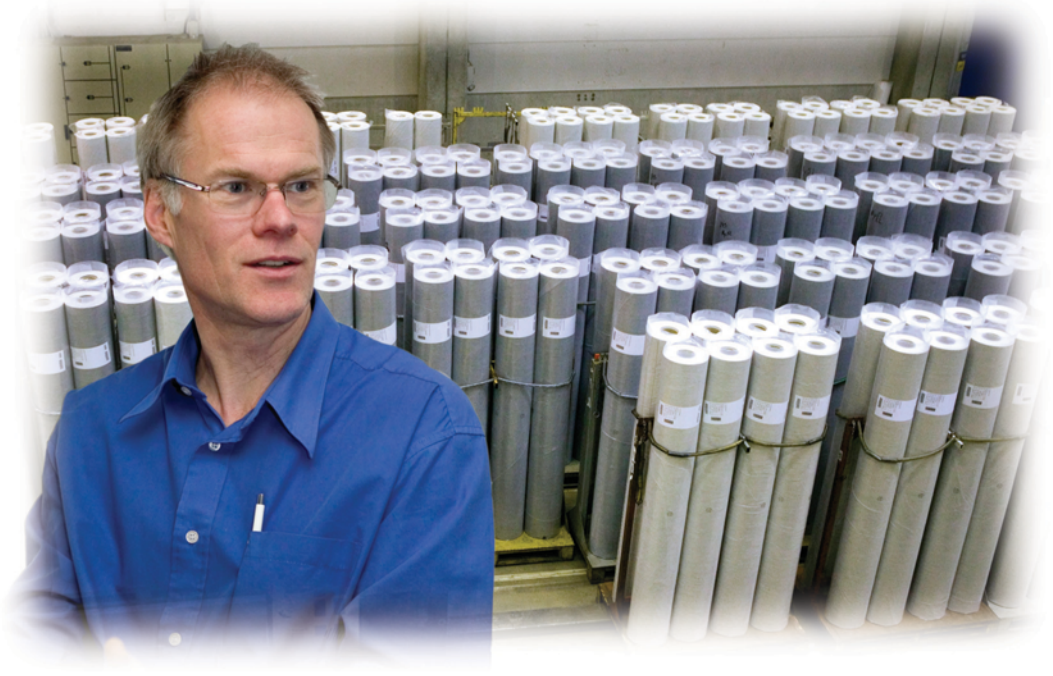
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—Jan Runnsjö, project manager, Tarkett.



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- Block instantiation, as specified in the capabilities file, has been added, insuring interoperability on a Foundation fieldbus network when a block is instantiated; and
- Testing for valid/invalid behavior during common software download has been added, to insure interoperability when downloading software/firmware to a device via a Foundation fieldbus network.

"The updated ITK includes expanded capabilities that allow testing of devices with DD 5.0 files. These DDs include new features focusing on device data organization, graphical visualization consistency, and support for persistent data storage," says Stephen Mitschke, FF's fieldbus products manager. "Today, at large plants with thousands of field devices, maintenance technicians must upgrade each device individually, using vendor-specific tools. ITK 5.0's common software download features allow instrument suppliers to update their device firmware 'over the wire.' Devices can be updated without needing a technician to go to each device and update a firmware chip or attach a custom upgrade cable. Upgrades enable suppliers to add additional diagnostic capabilities to 'future-proof' their existing devices."

Sensors Expo Turning 20, Will Feature 40 New Suppliers

More than 40 new exhibitors will highlight Sensors Expo & Conference, June 5-7, 2006, at the Donald E. Stephens Convention Center in Rosemont, Ill. This is the 20th anniversary of Sensors Expo, which is reportedly North America's leading event focusing on sensors and integrated systems.

"In addition to the high number of returning exhibitors, this year's event is attracting a large number of new exhibitors including Arched Rock Corp, Bielmatik, Dupont, IBM and Silicon Sensing," says Debra Brown, Sensors Expo's event director. "This widening interest reflects the excitement surrounding the revolution in sensing technology, as well as the quality of the overall program. The event brings together a community of sensor buyers, suppliers, researchers and educators."



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- Gentry Lee, chief engineer, Planetary Flight Systems Directorate at the Jet Propulsion Laboratory (JPL), will present "The Infinite Possibilities of the 21st Century" at 8:30 a.m. on June 6.
- Sebastian Thrun, associate professor of computer science and director of the Stanford Artificial Intelligence Laboratory (SAIL), will deliver "Winning the DARPA Grand Challenge: A Robot Race through the Mojave Desert" at 3:30 p.m. on June 6.
- Dr. Robert Metcalfe, Ethernet inventor and 3Com founder, will present "The Future of Embedded Networking" at 9:00 a.m. on June 7.

In addition, Sensors Expo's conference program is scheduled to focus on new innovations in sensor technology including physical sensors, sensor networks, biosensors, MEMS/nanotechnology, instrumentation and controls, intelligent systems, machine-to-machine communication, wireless sensing and IT technology.

For more information or to register, 800/748-5045 or visit www.sensorsexpo.com.

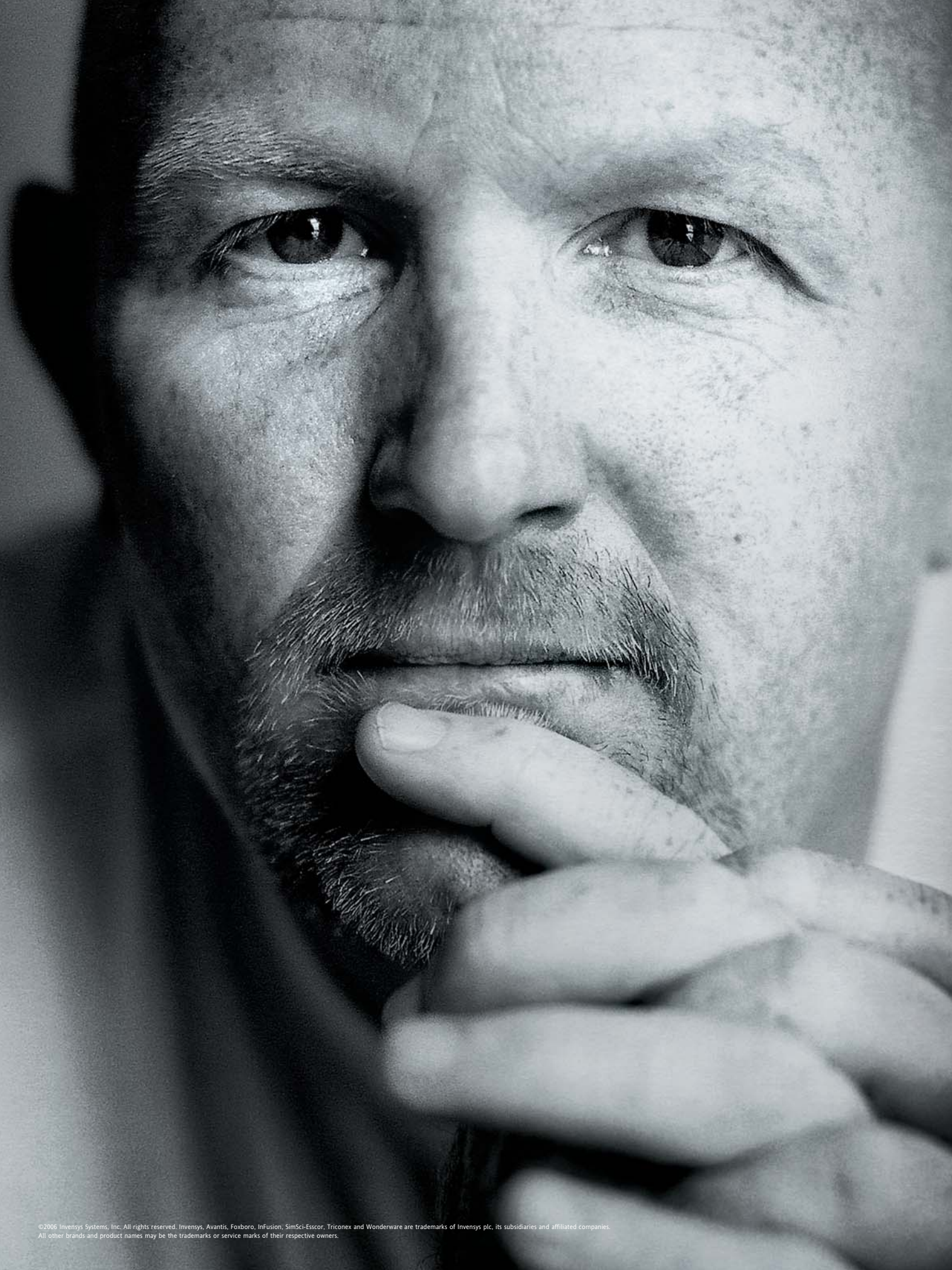
ISA Revises Equations Handbook, Publishes Test Measurement Book

ISA (www.isa.org) recently published a long-awaited, new version of its highly useful *ISA Handbook of Measurement Equations and Tables, 2nd Edition*, edited by Jim Strothman. This 314-page second edition follows the first edition, which was edited by William H. Cubberly, and published in 1994.

The new handbook provides essential equations and tables to help users select, operate, and maintain measurement devices. It also adds chapters packed with tables and equations for industrial communications buses, safety, and environmental measurements. Tables and equations have been added covering units of measurement, pressure, flow, temperature, level, humidity, electrical, and viscosity.

ISA also recently released *Fundamentals of Test Measurement Instrumentation* by Keith Cheatle. The new book is written for those who want to design test measurement systems; select appropriate equipment; understand system component characteristics, system and component calibration, and operating principles of transducers; determine overall system accuracy; and formulate basic test procedure design. The book also focuses on the practical application of test instrumentation and emphasizes the importance of creating a "measurement system" that involves components, installation, wiring, and calibration.

For more information, visit www.isa.org/books.

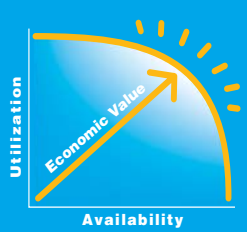


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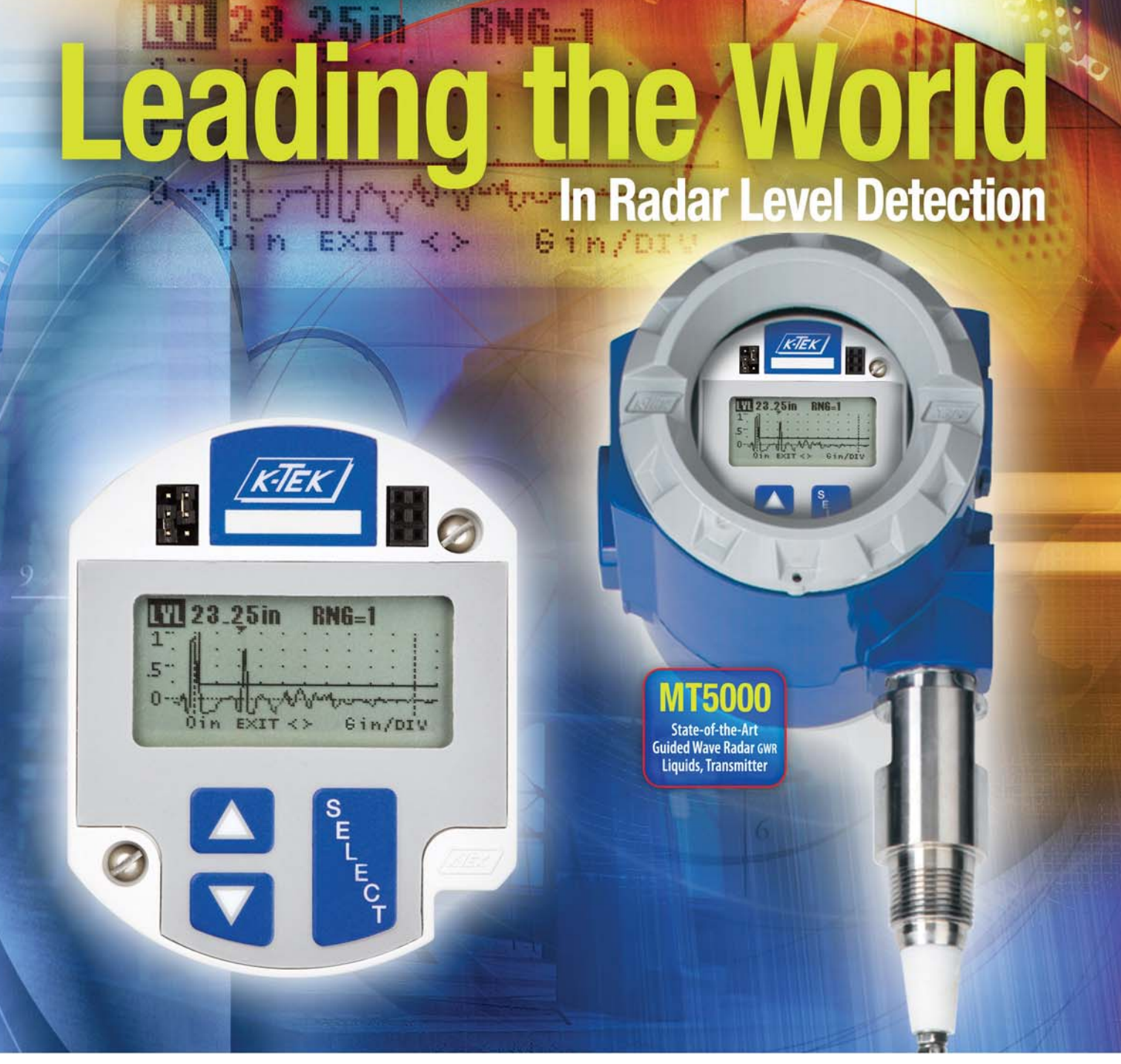
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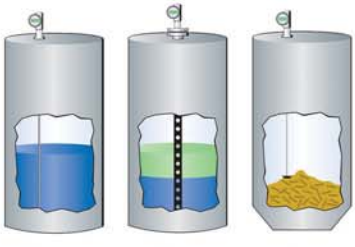
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Nuclear Power Makes Sense

We've covered this topic before, but recent events make it more important than ever. We need to build hundreds of new nuclear power plants. This is true, whether or not you believe we can do something about global warming, or whether or not you think the world's oil and gas reserves are running out, or whether or not you believe we need to start using electric-powered cars.

Most of our oil is imported, which makes us vulnerable to terrorists, who can overthrow governments, blow up pipelines, and otherwise cut off our supplies. In Germany, where the Green movement made politicians cave in, and start closing nuclear plants, this is a real problem. Germany relies heavily on Russian oil and gas, and Russia isn't exactly stable, so the planned phase-out of nuclear reactors that supply one-third of Germany's electricity is being reconsidered.



Building nuclear plants will create thousands of jobs for control engineers, operators, and technicians.

Global warming enters into the decision to go nuclear, especially in Europe. Paris-based *Technology Review* writer Peter Fairley recently discussed the thinking about nuclear power in Europe with Lars Josefsson, CEO of Vattenfall AB, a leading producer of electricity and operator of nine reactors in Sweden and Germany.

"Finland is building a nuclear power plant, and France looks set to follow suit with one of its own," says Fairley. "Do you expect other countries in Europe to join the trend?"

"It's quite a process to decide to build new nuclear," says Josefsson, "and one that will take several years. But the fact that there is a trend shift in Europe is, to me, obvious. Take Britain. It's moving in that direction very clearly. And I think the replacement market for aging plants in Europe will be sizable."

European countries are under pressure to live up to their commitments to the Kyoto Protocol, which calls for a decrease in emissions. To many Europeans, nuclear power is the only way to accomplish this.

In the U.S., most new power plants built in the last 25 years have been gas-fired. Natural gas is getting to be downright expensive, so power plants are looking to coal as an alternate power source. The McIlvaine Co. (www.mcilvainecompany.com), a market research firm, says $48 billion per year will be invested in new coal-fired boilers over the next 10 years.

But coal has its problems, too, including acid rain, damage to land from mining, medical costs, and, according to writer Bruce Sterling, "Coal spews more weather-wrecking pollutants into the air per unit of energy than any other fossil fuel." (*MIT Technology Review*, Oct. '03).

Ethanol shows promise, especially as a fuel for motor vehicles, but it can't replace all the imported oil. Therefore, many people are calling for increased use of electric cars. Problem is, charging up all those vehicles' batteries every night will require more electric power generation.

Nuclear power makes more sense than anything else. A new nuclear plant costs as much as a new coal plant (about $1 billion dollars, more or less). It produces no global warming-type emissions. It requires no imported hydrocarbon fuel. Siting and licensing problems are easily solved.

Put them on abandoned military bases, for example. Those bases have plenty of security, are owned by the government, so there are no (not in my backyard) NIMBY problems, and the surrounding populace would probably welcome them as a source of construction jobs.

As for licensing, I've said this before: get rid of the old fogies running the Nuclear Regulatory Commission (NRC), put in some of *Control*'s readers, and adopt a universal design for the instrumentation and controls. If Westinghouse and Babcock & Wilcox build 'em the same way Europe does, nukes won't take 20 years to get approval.

Because of the NRC's stifling rules, most control and instrumentation vendors in the U.S. abandoned the nuclear business years ago. Josefsson says, "There's a real risk that the nuclear technology supply industry will become a bottleneck in the near future." If the NRC adopted more reasonable rules, it could give our control equipment vendors a tremendous shot in the arm, especially if the U.S. was to build 50 new nuke plants per year for the next 20 years.

Building nuclear plants will also create thousands of jobs for control engineers, operators, and technicians. The best part is the $1 trillion dollars that will be spent on building 1,000 nukes will stay here, in the U.S., where it will benefit us, not some third-world country.

Rich Merritt, senior technical editor
rmerritt@putman.net

Focusing on Fieldbuses

***CONTROL*'S Monthly Resource Guide**

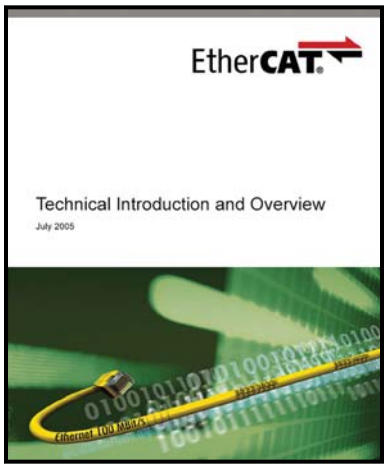
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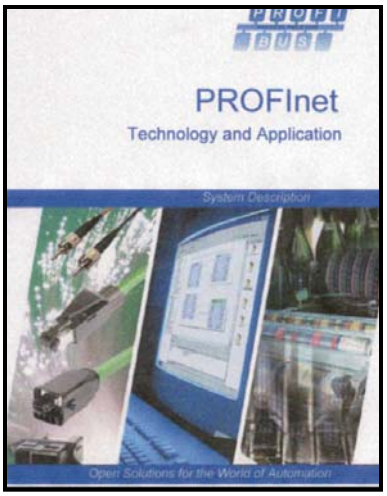
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FOUNDATION FIELDBUS SITE

The official site of Fieldbus Foundation contains support materials and services including a registered product catalog, newsletter and a Host Interoperability Support Test (HIST) results page that lists all Foundation fieldbus-compliant host systems tested for functionality defined by the foundation's host feature checklist. Application guides describe using FF for industrial process automation, and end user presentations, white papers and technical articles discuss FF projects worldwide. **Fieldbus Foundation; 512/794-8890; www.fieldbus.org**

FOUNDATION FIELDBUS BOOK

The second edition of *Foundation Fieldbus* by Ian Verhappen and Augusto Pereira includes rules for cabling length, documentation requirements, a commissioning checklist, topology diagrams, system sizing formulas, and tips for integrating with other systems. It explains the Fieldbus Intrinsic Safety Concept (FISCO) along with configuration and troubleshooting tips. **ISA; 919/990-9287; www.isa.org**

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The website below is dedicated to fieldbus solutions, products, technology, educational resource materials, downloadable literature and whitepapers. The site contains links to related standards organizations, industry associations, higher-education facilities, and solutions providers. An "Ask the Expert" section promises a response within one business day. It also has a Segment Checker open software tool that supports centralized and distributed fieldbus architecture concepts for Foundation fieldbus and Profibus PA. **Pepperl+Fuchs; 330/486-0002; www.pfsolutions.info/FieldConnex/**



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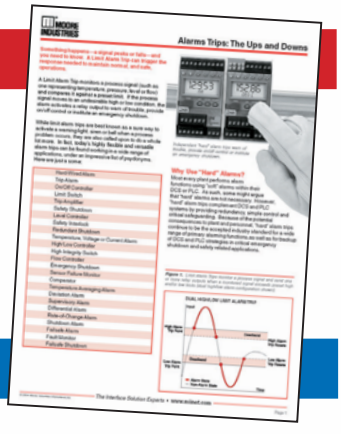
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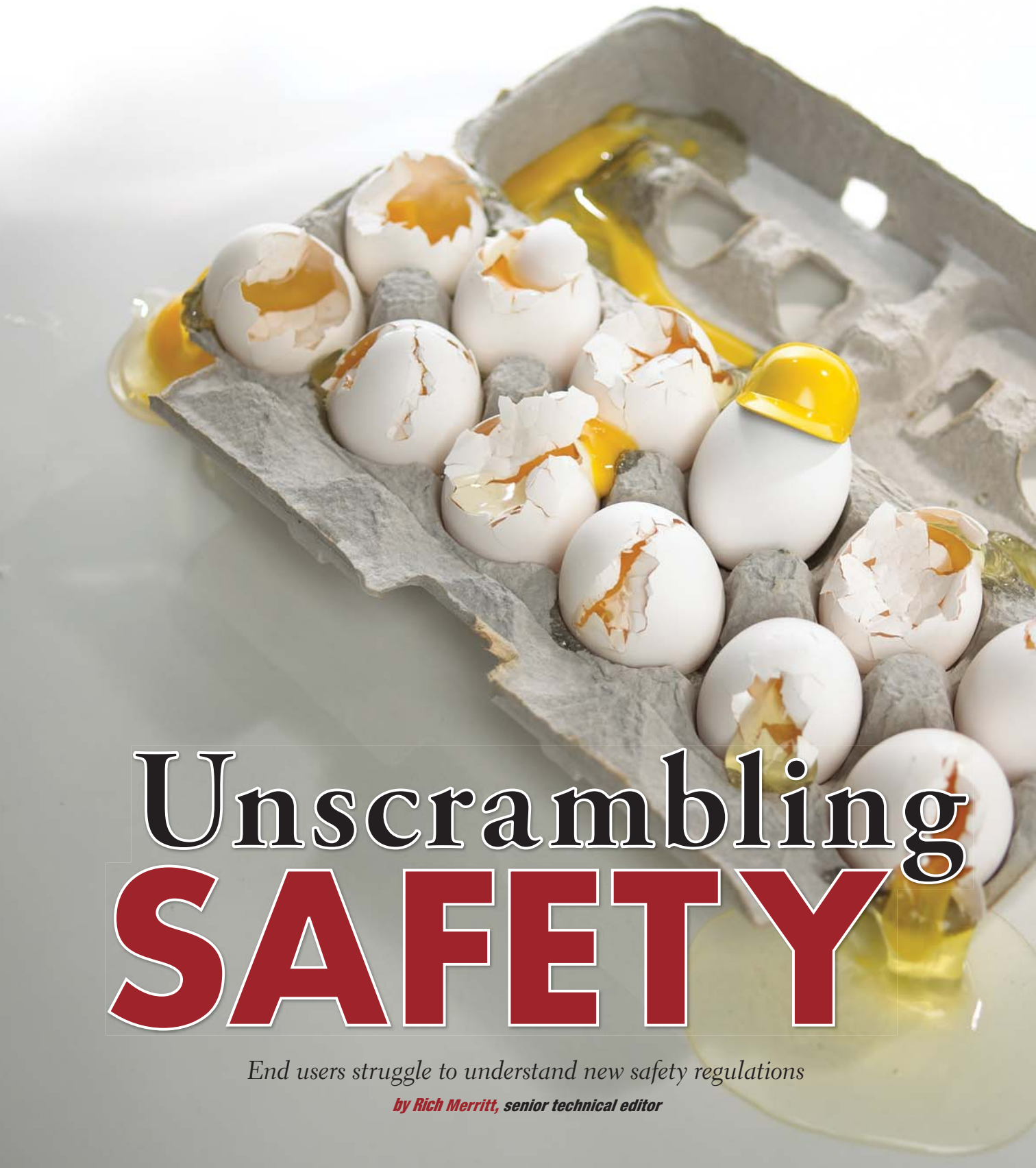
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Unscrambling SAFETY

End users struggle to understand new safety regulations

by Rich Merritt, senior technical editor

Some users are embracing safety standards, some are confused by the terminology, some just don't understand it at all, and some are afraid to talk about it in public. Many just turn safety issues over to process control vendors, and some work hand-in-hand with outside experts (Figure 1). In this article, we'll look at how some users and vendors are dealing with safety issues.





Making Stuff Safely

Eric Marcelo, supervisor at the Nestle Philippines coffee factory in Cagayan de Oro City, Philippines, says his plant has a positive attitude toward safety. He even understands all the safety requirements that apply to his plant. "We were given training at the start of our employment and whenever there are changes in standards or installations," he says. "Basically speaking, safety is everyone's concern. We have a safety officer for the whole factory, safety champions for each department, and safety experts for certain processes. The safety expert ensures that safety procedures, interlocks and safety features of equipment and processes are followed or adhered to. Any changes will have to be cleared through him, and will undergo complete analysis before permission to proceed is given. In certain situations, any suggestions or problems he receives, or is made aware of, will be sent to the R&D center for more thorough analysis and testing."

What's more, management supports these efforts wholeheartedly. "Safety is everyone's responsibility, whether it's his own, or that of his fellow workers/consumers," adds Marcelo. "Management sees to it that safety rules and procedures are followed. We comply with government regulations with regular inspections and periodic tests. We follow electrical and mechanical standards, and we conduct regular safety inspections and internal audits."

Some users turn to automation vendors to solve their safety issues. For example, the Rompetrol Rafinare refinery and chemical complex, located on the Black Sea in Navodari, Constanta, Romania, recently installed two safety instrumented systems (SISs).

Cristian Pariza, Rompetrol's automation/systems engineer, managed the installation of SIL 3-rated DeltaV SIS technology from Emerson Process Management (www.emersonprocess.com) to protect the refinery's gas-fired burners for atmospheric distillation/vacuum distillation (AD/VD) heaters. In addition, SIL 3-rated SIS equipment was placed on polypropylene plastic and pyrolysis units in the facility's petrochemical plant. Configuration, startup, and post-startup activities were conducted by a team of refinery and Emerson engineers.

"Looking at the AD/VD project in particular, the safety system consists of three identical subsystems, one for each heater," says Pariza. "Each subsystem is housed in a dedicated cabinet (Figure 2) containing 18 logic solvers, 270 I/O points, redundant power supplies, and redundant communications with the plant network.

"A great advantage of the Emerson SIS is its integration into the plant's existing DeltaV hardware, software, and global database. No additional operator, application, or engineering workstations were required. What's more, the safety system's operating style is the same, it uses the same configuration and diagnostic tools, and display screens are similar."

Meanwhile, the Rashtriya Chemicals and Fertilizers plant in Chembur, Mumbai, India, manufactures 900 metric tons per day of ammonia, plus quantities of urea, nitric acid, and other chemicals. When its engineers wanted to update Rashtriya's mostly pneumatic and relay-based control systems, they decided to do the process controls and safety systems all at the same time, using an integrated system from Honeywell (Figure 3).

This new system includes an Experion PKS control system and a SIL-3 rated Safety Manager, which focuses on safety-related process variables, and prevents unacceptable access or interference from operators or maintenance personnel. Safety Manager also is used for compressor-process interlocks, and initiates an emergency shutdown if unhealthy process values are detected. This prevents unsafe shutdown practices, which can cause injuries or damage plant equipment. It also limits nuisance trips.

Similar to Rashtriya and Rompetrol's safety equipment, Triconex's Trident system integrates process controls and safety at oil and gas wells for the Al Noor and Al Shomu oil fields in the desert of South Oman. The oil wells are unmanned, and operate continuously. The main function of the Trident PLC at the wellheads (Figure 4) is to act as an instrumented protective system for the pipeline to a production station, the wellhead, and the oil reservoir. Trident's triplicated architecture helps maintain production of the well from a single fault or processing error. If a hardware fault occurs, the module can be replaced online, so the well can keep operating and flowing at all times.

Teams Refine Answers

As you might expect, some users in refineries are somewhat reluctant (or prohibited) from talking about safety issues. Two end users—Ed and Bob—described their company's efforts, but have to remain anonymous.

Ed claims to understand all the safety requirements, as far as control systems, process equipment and Safety-Instrumented Systems (SIS) are concerned.

"Within this sphere, I'm very knowledgeable about the design, maintenance, and operation of safety-instrumented systems," says Ed. "This doesn't imply I know it all, but rather that I am competent enough to lead and oversee design efforts."

Ed reached that point by leading his plant's safety effort. "I participated in the early phases of developing our internal SIS standard and application practices, and most recently in the design and application of several systems. While I can't claim to know it all, experience and knowing who else in and outside the company to call for answers to certain questions does help."

Refineries are big places with lots of equipment, and no one can be expected to know all their safety regulations. "My knowledge is limited to the design of systems and the intent of our standards. It doesn't extend to the specific process hazards of any given process. No control engineer could be reasonably expected to understand

FIGURE 1.

TRANSITIONS

The sunset of old IEC safety regulations and the dawning of new SIS and SIL rules make safety systems a real challenge for control engineers, especially at large installations, such as the Rompetrol Rafinare refinery and chemical complex in Romania.
Source: Rompetrol Rafinare



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all the hazards of the processes in his facility, nor should he advertise or assume that he does."

Like Marcelo's coffee plant, Ed and Bob's refinery takes a team approach to safety. "The team determines where safety systems need to be applied and what their design should be," he explains. "The team is run by a project engineer, who is knowledgeable about safety system design. Other team members include Operations, Engineering, Instrumentation, and Maintenance departments. We use outside consultants to facilitate the design meetings, develop the detailed design, and implement the system in cooperation with the control group. For design, it's best to involve external consultants, who are experts at SIS design and implementation. We also have corporate support on safety system design and process technologies.

Because refineries are subject to many rules, Ed's team has to meet ISA S84, API-521, API-556 and NFPA-85 specs. They also use Triconex's Trident to meet SIS requirements. However, the refinery goes beyond the rules. "While ISA S-84 is a performance-based standard, and this is a good thing, it's not enough," he says. "Our internal standards and design practices are more prescriptive."

Our other refinery engineer, Bob, takes a similar approach. "We have a fire and safety manager, a PSM coordinator for OSHA 1910 Process Safety Management, and a corporate charter related to safety," he says. "The fire and safety manager has all the responsibilities and authority of being a manager of a department that has corporate scrutiny. The PSM coordinator is primarily responsible for paper."

Bob explains that safety requirements are necessarily widespread in a refinery. "There's process safety related to relief valves, vessel pressure ratings, design and SIS," he notes. "Also, there's personnel safety, such as clothing, training, ergonomic designs, and injury reduction. Then there's infrastructure safety, including electrical safety and physical property protection, such as fences, barriers, lighting, cameras, and guards."

Because of its complexity and scope, safety is an ongoing task. "We use Triconex PLCs to meet our basic practice specification which is based on S84. We're currently revising our Basic Practice to meet the revised S84, which follows the IEC specification," adds Bob. "We follow the safety lifecycle in our design phase, do SIL reviews, develop SRSs, and do everything advised by our safety consulting firm. We require our primary engineering contractor to work with our designated safety consultant to complete designs. We presently have safety systems on all of our processes, with one-third meeting the S84 standard, but with plans to convert all the plants on a scheduled basis. We have active projects to convert six plants this year."

Sven Grone, president of HIMA Americas Inc. (www.hima-americas.com) in Houston, Tex., adds that safety systems are becoming more complex-because many facilties are adding support utilities and other formerly peripheral applications to their overall safety infrastructures to comply with S84.

"In the past, safety systems were concentrated in the main production processes, while the utility boilers and process heaters around the plant used mostly relays and PLCs," says Grone. "Now, uses are having to reevaluate if they need safety systems in support areas. Users are also having to evaluate their existing processes and assign SIL levels. A process should implement a certified safety systems if it's judged to be above SIL 0."

What You Don't Know

Understanding all the aspects of safety is a daunting task. Mike Reilly, an engineer at Flint Hills Resources, a refining and chemical company in Wichita, Kan., says he understands his safety rules…he thinks. "But you don't know what you don't know," says Reilly.

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In a recent survey of *Control*'s readers, we asked: "Do you understand all the safety requirements that apply to your plant, such as ISA S84, SIL, PSM, ESD, etc.?" More than half the respondents answered "no."

Chris Conklin, senior engineering specialist at Dow Corning Corp. in Midland, Mich., says he too doesn't understand all of his safety requirements. "We went through a major workforce reduction a few years ago, and we lost significant knowledge regarding instrumentation, process controls, and safety instrumented systems," he explains. "So, to answer the question honestly, I would have to say we don't understand all of the safety requirements that apply to our manufacturing site with respect to safety instrumented systems."

So who does? "Basic safety responsibility and authority currently resides in the safety and loss control department," adds Conklin. "However, as far as I know, no one in that group has the knowledge or understanding about the safety requirements with respect to S84, SIL, ESD, etc. In our group, we recently identified this as an issue, and we're seeking someone who can become our subject-matter expert in safety-instrumented systems."

Len Laskowski, Emerson's technical consultant and engineering fellow, adds, "Today nearly all companies are experiencing the lack of staffing to deal with all the new regulations, standards, and other issues. Some companies with long-established guidelines dealing with SIS systems are updating their internal standards to stay current with international standards such as IEC 61511. Others, who had no or minimal guidelines, find it easier to just adopt the new standard. Some companies are still trying to assess the new standards, and determine what they must do to be compliant, and there are a few that have yet to do anything."

Louis Szabo, business development manager at Pepperl+Fuchs (www.us.pepperl-fuchs.com), says end users are confused about safety requirements. "For instance, in one case, a customer used a 'rule of thumb' they obtained from Exida that provided parameters for a generic IS barrier," he says. "Their understanding of MTBF data was incorrect, as was their understanding of SIL 1 through SIL 4. To them, MTBF meant 'Mean Time Before Failures,' instead of the actual 'Mean Time Between Failures' definition. SIL 1 meant one year, SIL 2 meant 10 years, SIL 3 meant 100 years, and SIL 4 meant 1,000 years. This misconception was cleared up, and they're now safer for it. In another case, a customer's safety system was never tested. They experienced a failure in the primary control

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system, and the back-up never triggered, resulting in a chemical spill and a $10,000 EPA fine."

Charles Fialkowski, product manager at Siemens Energy & Automation (www.siemens.com) has similar horror stories. "I've heard some major oil and gas companies say, 'Most here don't know how to spell SIL,' which is rather scary. While I don't question their corporate knowledge, it's usually the local plants that are hurting the most for information and knowledge. From my personal perspective, over the past 10 years, I've seen considerable increase in knowledge and awareness across the board, which is very encouraging."

Read the Manuals!

Unfortunately, improving safety is complicating by safety's own standards nomenclature. Laskowski explains, "Just for the record, ANSI/ISA 84.00.01-2004 Part 1 is really IEC 61511-1, with the exception of clause 1y, the grandfather clause. This is frequently referred to as S84 in SIS circles. This standard doesn't govern development of equipment to be used in SISs, nor does OSHA's STD 29CFR1910.119, commonly referred to as PSM. Instead, Emerson and other vendors use IEC 61508 as the benchmark to develop hardware and software for equipment that will be SIL rated and then TÜV certified as acceptable to use in SIL-rated applications." No wonder so few users understand safety issues.

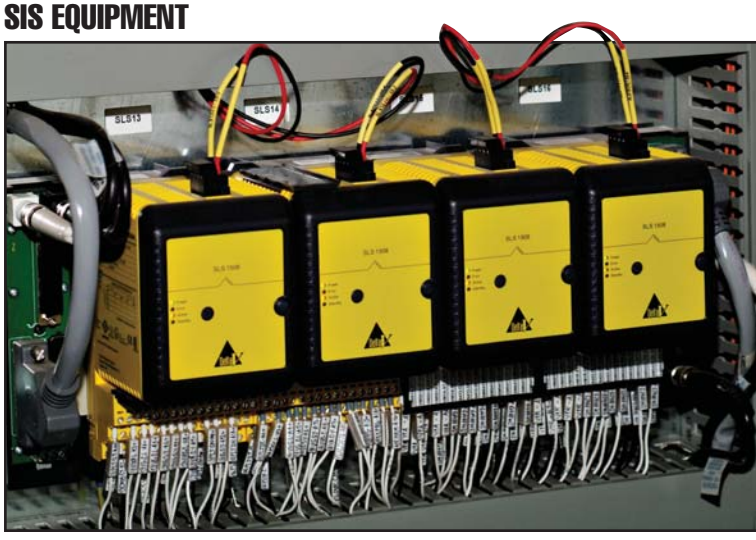
Scott Hillman, safety management systems manager at Honeywell (www.honeywell.com), is TÜV-certified, so he appreciates how tough it is to get up to date. "Users are taking safety rules very seriously, and are earnestly attempting to meet these requirements," he says. "They're struggling with interpreting the standards, how to apply them, and what resources they have to implement them. A few end-users have the capability to respond internally. Most customers, however, will make use of the safety expertise available at Honeywell or other supplier organizations to implement their safety requirements into a safe, available, user-oriented safety solution. They realize that safety isn't just a regulation they have to meet, and that it's good business to operate their plant safely."

The cost of safety is often misunderstood, too. Connie Chick, manager of the PACSystems group at GE Fanuc (www.gefanuc.com), says users are confused. "They are getting confused a lot of times by our own industry, which sometimes uses safety as a selling point. The customer doesn't understand what a 'total installation' requires. For instance, a SIL 3 triplicate system can involve triplicated everything—I/O, devices, etc.—which then becomes really expensive. Jumping to that level is a real cost hit, and you need to understand whether it's truly required for the application."

If you want to do it yourself, Hillman has some advice. "End-users must read the applicable manuals to determine how to implement the system to meet the SIL 3 certification. It's a 'buyer-beware' scenario. They must realize that TÜV certificates are one thing, but the information in the safety, installation, and implementation manuals dictate how the system must be implemented to meet the SIL level. These describe the restrictions and requirements to implement to the SIL level. Those restrictions often represent additional configuration and cost to the end-user.

FIGURE 2.

SIS EQUIPMENT



The DeltaV safety systems at Rompetrol are housed in a dedicated cabinet containing logic solvers, I/O points, redundant power supplies, and redundant communications.
Source: Rompetrol Rafinare

Lakowski adds, "While we see the entire spectrum of experience, when it comes to dealing with SIS, most customers are aware of the standards, but haven't fully come to appreciate the requirements. Since IEC61511 is a performance standard, measures of performance are required. The good news/bad news is that some engineering is needed to develop the performance measures. In reality, there's more work for engineers to do to implement this IEC standard than was the case in the old days of 'cookbook' or prescriptive standards. In every proposal that we send out involving smart SIS with DeltaV SIS, we break out the steps of work in the IEC61511 Safety Life Cycle. There are more than 40 steps we recognize in the portion of the lifecycle normally associated with projects."

Vendors to the Rescue

For users who don't know the difference between a MTBF and SIL 3, process control system vendors offer plenty of help. They will even take on the entire job. Simon Coleman, Clair project manager for British Petroleum, selected Honeywell to install an Integrated Control and Safety System for its Clair offshore platform (Figure 5), located 75 kilometers west of the U.K.'s Shetland Isles.



"Honeywell engineers participated in SIL Safety Integrity studies to categorize the safety, environmental and commercial integrity level for every aspect of the platform," explains Coleman. "Honeywell's fire and gas engineers were involved with the EPC contractor and the Topsides 3D model to locate gas detectors, smoke and heat detectors, and fire and gas closed-circuit TV cameras. Abnormal Situation Management (ASM) standards were used in generating the HMI to minimize information displayed to the operator to safely operate the platform. This included an alarm minimization review, in which every alarm in the Clair system was reviewed, and uniquely identified with priority, cause, and remedial action."

Coleman says working with one vendor is a huge benefit. "We don't have two vendors supplying two different interfaces. This helps us avoid unnecessary communications, and makes graphics and displays consistent."

Likewise, a systems integrator can help with installation. "We have folks that are trained in the workings and specifications of SIS systems, but we don't engage in defining these systems, only in executing the plans of the owners," says Cliff Speedy, project engineer at C&I Engineering in Louisville, Ky. "We'll participate in the planning of the systems, but most of the real definition comes either from experts from the corporation or from hired SIS consultants. We've become real experts at estimating and installing these systems, rather than experts at defining them."

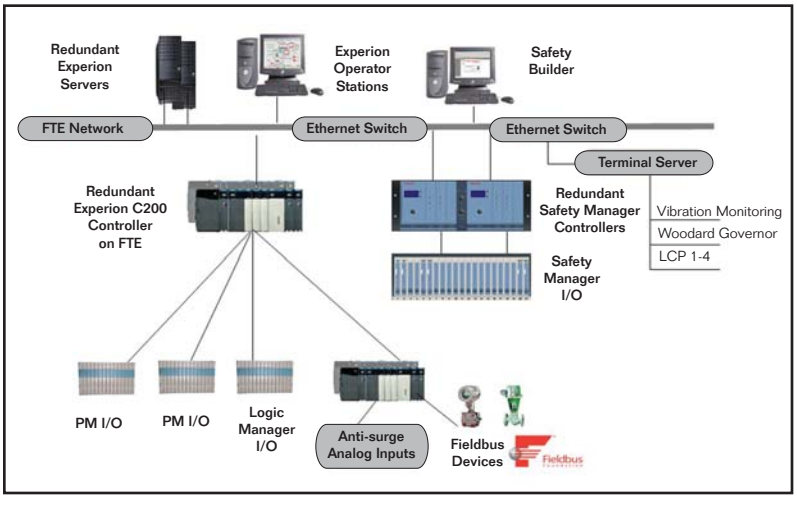
Hillman adds that Honeywell also provides services that cover the entire safety lifecycle, or they will help you do it yourself. "Our project engineers and our integration centers are certified to the TÜV international standards," he says. "We're approved by TÜV to train and certify to the safety standards, so end-users can design and install themselves. We host training classes, and we're listed on the TÜV site as certified trainers."

Siemens does the same. "We offer advanced tools to assist the end user thru all critical phases of the safety lifecycle," says Fialkowski. "We can provide front-end engineering services for HAZOP and SIL validations, we have 'Centers of Excellence' for the oil and gas and chemical industries, and we offer pre-certified safety designs in applications like burner management systems and fire and gas."

Ged Farnaby, North American business development manager for

FIGURE 3.

FERTILIZER SAFETY



The safety system for a chemicals and fertilizer plant is integrated with the process control system. Source: Honeywell.

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the Safety Solutions Group of ABB (www.abb.com) says his company can help, too. "ABB has made it a priority to help customers with safety system implementations, and to ensure our internal compliance for these solutions," he says. "We have numerous certified safety engineers worldwide, as well as dedicated Centers of Excellence for safety. We've worked with our quality and engineering groups to produce a set of guidelines and procedures to ensure that all safety systems are designed and delivered for full compliance to ANSI/ISA 84.00.01 2004 Part 1-3 (IEC 61511-3 Mod)."

In addition, Triconex and Invensys work together to help users. Triconex and Invensys reportedly provide TÜV-certified engineering resources worldwide, which assist in the design, programming, implementation and installation of safety instrumented systems. Beyond a stringent quality assurance scheme and best engineering practices, providing qualified personnel in safety matters is particularly important as experienced plant personnel are getting scarce."

Laskowski adds, "In addition to classroom, on-site training, and e-learning courses, Emerson offers courses on SIS at its PlantWeb University online learning center. The courses provide fundamentals and practical tips for SIS planning, selection and implementation, plus information on basic SIS concepts, design and installation, operation and maintenance, safety standard compliance, and new SIS technologies."

It will certainly be beneficial to learn as much as you can about SIL, SIS and all the other safety regulations. Turning it all over to your process control vendor, while easy, won't save you.

FIGURE 4.

SINGLE-VENDOR SOLUTION



Honeywell installed an integrated control and safety system for British Petroleum's Clair offshore platform.
Source: BP



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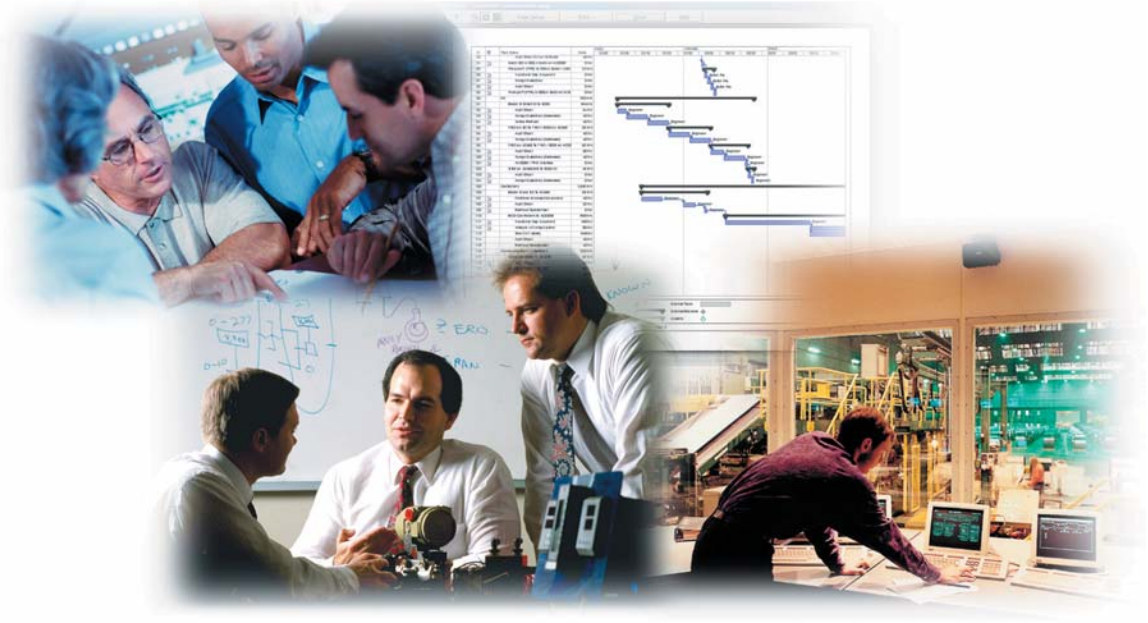


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THE POWER OF EXTERNAL-RESET FEEDBACK

External reset will save your process from the curse of windup

by F. Greg Shinskey, process control consultant

The integral mode of a controller is essential in eliminating offset in any loop subject to load changes—which is almost every loop encountered in process control. Consequently, almost every controller has integral action in one form or another.

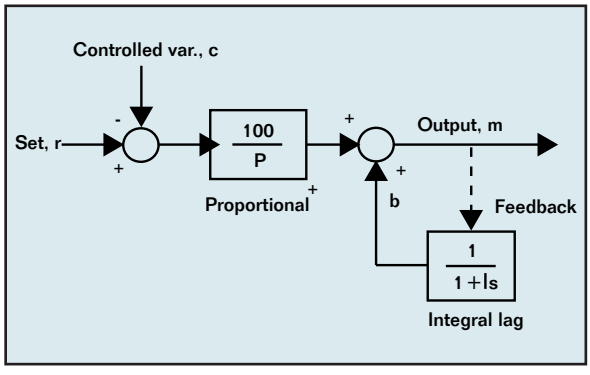
However, tireless effort to drive deviation between the controlled variable and its set point to zero presents problems when the loop is open. In an open loop, no amount of control effort will be successful, and continuing to integrate will result in "windup" with adverse consequences. The integral component in a "wound-up" controller doesn't balance the process load, driving the controlled variable away from a steady state when the loop is eventually closed. This imbalance can produce a large deviation, requiring integral action to eliminate, and, if repeated, it results in a limit-cycle immune to correction by tuning the controller. Preventing windup requires reconfiguration of the controller, and external-reset feedback is the most satisfactory method of accomplishing it.

Development of Automatic Reset

The first pneumatic controllers had only on-off action. A mechanical linkage, whose position represented the difference between process measurement and set point, acted on a relay to switch the output pressure between its two states. Proportional control was introduced by negative feedback of that pressure to reposition the linkage through a bellows.

FIGURE 1.

AUTOMATING RESET



Automatic reset is achieved by positive feedback of the controller output.

The precise value of the output when the deviation was zero could be adjusted by a screw, acting on a spring opposing the bellows, as a bias (Equation 1):

$$m = \pm\frac{100}{P}(r - c) + b$$

in which m is the controller output, c and r are the controlled variable and set point, P is the proportional band, and b is the bias, with all variables expressed in percent of scale.

Proportional offset develops whenever the process load requires a value of controller output that isn't equal to the bias. Consider, for example, a proportional-level controller with a 50% bias manipulating the flow entering a tank. If the flow leaving that tank exactly matches the flow entering with the valve 50% open, there will be no offset. However, at any other value of outflow—which requires a matching inflow to reach a steady state—the matching inflow can only be attained by a proportional offset.

The idea came first, and theory followed. The correlation between automatic reset and integral action wasn't commonly called by the name "integral" until 1970.

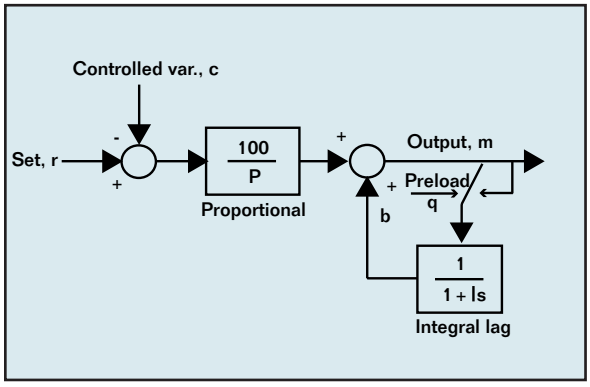
In applications where offset was particularly undesirable, a plant operator might reset the set point to position the controlled variable where it was wanted. The offset remained, however, and it was variable. It could be manually eliminated by operator adjustment of the output bias, and this became known as manual reset, but only until the load changed again.

In 1929, "Doc" Mason of the Foxboro Co. came up with the idea of replacing the bias spring with a bellows connected to the controller output. If the bias and controller output could be kept equal in the steady state, Equation 1 shows the offset will be zero. This became automatic reset.

However, simply connecting the two bellows together produces an on-off controller, as the new bellows adds positive feedback to the controller, canceling the negative feedback

FIGURE 2.

BREAKING THE PATH



The batch switch closes the reset-feedback loop when output is below 100%.

of the proportional bellows. To stabilize this loop, the positive feedback has to be slower than the negative feedback coming from the process. So, a restrictor was inserted between the controller output and the feedback bellows, creating a first-order lag.

Initially the restrictor was fixed, then a selection of fixed restrictors was used, and finally an adjustable restrictor. This configuration is shown in Figure 1, in which time constant "I" was known as the reset time.

This is one of those examples where the idea came first and theory followed. Only much later were equations used to develop the correlation between automatic reset and integral action, and it wasn't commonly called by the name "integral" until 1970.

Substituting controller output m for b in Equation 1, and applying first-order lag I gives (Equation 2):

$$m = \pm\frac{100}{P}(r - c) + \frac{m}{1+Is}$$

where s is the Laplace operator.

Rearranged, we have (Equation 3):

$$m = \pm\frac{100}{P}(r - c)\left(1+\frac{1}{Is}\right)$$

which is recognizable as the proportional-plus-integral controller algorithm.

The reset feedback path in Figure 1 is shown as a dashed line, indicating that it can be broken to stop integration, and prevent windup for reasons explained in the various applications below.

Not all controllers integrate by means of a feedback loop, however, and these require other methods of windup protection that aren't as effective. So, it may be necessary to build a controller from the elementary function blocks shown in Figure 1 to obtain the capability of external-reset feedback.

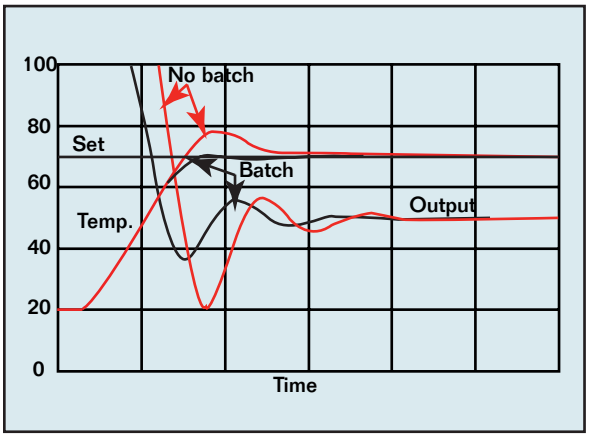
Batch Control

An early problem caused by integral windup was temperature overshoot in heating batch reactors. The reactor was charged cold, with the temperature controller in automatic, set point at its desired value, and the steam block valve closed. Opening the block valve started the heating operation. At this point, the reset bellows contained full-supply pressure, keeping the controller output saturated and the steam control valve wide open until temperature crossed its set point. The resulting overshoot was unacceptable. The remedy shown in Figure 2 breaks the reset-feedback path with a batch switch whenever the controller output exceeds 100% or wherever the manipulated variable limits, and substitutes a manual-loading signal. Suitable adjustment of this preload positions the integral term near the anticipated process load, which can prevent overshoot, and provides a smooth transition from proportional or proportional-plus-derivative to PI or PID control. This same method is now applied to digital controllers with external-reset capability.

Figure 3 shows how an unprotected (no-batch) PID controller can cause temperature overshoot during startup of an inert batch when the integral term is saturated at 100%. Derivative acting on the controlled variable causes the output to leave its limit before set point is crossed, but not soon enough to avoid overshoot. In an exothermic reaction, the overshoot would be even more severe, and could result in product loss. The controller protected by a batch switch is preloaded near the anticipated load of 50%, where split-range heating and cooling valves are both closed, thereby avoiding the overshoot. The controller's mode settings can be tuned for optimum load rejection, and the preload adjusted to produce the desired approach to set point on startup. Too low a setting results in undershoot, and too high a setting causes overshoot. A preload setting of 100% would produce the same overshoot as the no-batch controller in Figure 3.

FIGURE 3.

AVOIDING OVERSHOOT



Proper preloading of the batch switch optimizes the approach to set point.

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While the loop is open, if the deviation happens to lie within the proportional band of the controller, its output may fall below the switch setting after transfer to preload. Then, the switch will reverse position, and integration will resume, driving the output back to the switch setting. The switch will then cycle between its two positions.

In a digital controller, this reversal could happen at every scan interval. The ultimate effect is essentially to control the controller's output at the switch setting, with the integral term "b" settling at an intermediate position representative of the current process load.

Any upset reducing the deviation will then be countered by an immediate change in controller output away from the limit. This behavior is common in anti-surge compressor control, where suction flow is normally well above the surge set point and the recycle valve is held closed by the flow controller. A sudden loss in load will reduce flow, and the controller must start opening the recycle valve before the set point is reached.

The batch switch applies equally well when the manipulated variable encounters a lower limit. The switch then transfers to preload when the controller output falls below that lower limit.

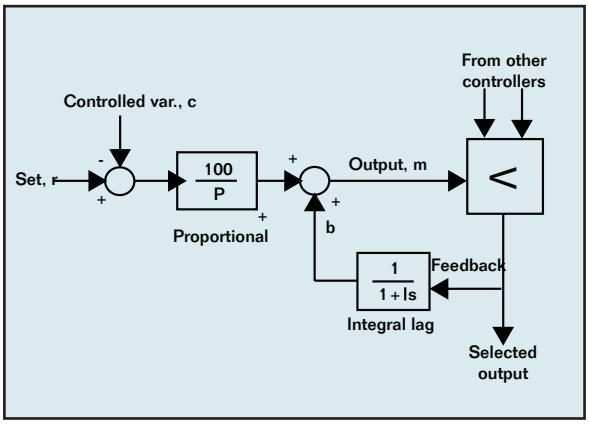
Override Control

Override control was recently described recently in H. L. Wade's "Under the Hood of Override Control, Parts 1 and 2," *Control*, Dec. '05, pp. 43-45, and Jan. '06, pp. 59-61. In these systems, two or more controllers compete for the same manipulated variable, which is selected based on whichever has the lower or higher output. Since only one controller can be selected at any given time, the remaining loops are open, and their unselected controllers will wind up unless protected.

Figure 4 shows the most effective method for avoiding windup in the unselected controllers: the selected output

FIGURE 4.

PREVENTING WINDUP



The selected output is fed back to the integral term of all controllers.

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is the common feedback signal to all of the controllers in the system. Simply stopping integration in the unselected controllers isn't enough because it leaves their integral term loaded with a constant that loses its relationship to process load over time.

The selected output, however, represents current process load, and therefore keeps the unselected controllers current while their loops are open. Only the selected controller sees its own output fed back, and so it alone integrates. Because all controllers are biased at the same level, transferring control from one to another will tend to take place when both are at zero deviation, and it therefore becomes a smooth event.

The integral lag plays an important role in this transfer. Other methods of windup protection tend to be too abrupt because they include forcing all controller output limits to track the selected output, or periodically initializing the unselected controllers. Any noise on the selected output is easily rectified into a bias with these methods, potentially shifting the transfer point on the unselected controllers, or causing offset in the selected controller. The integral lag avoids this problem by effectively filtering noise.

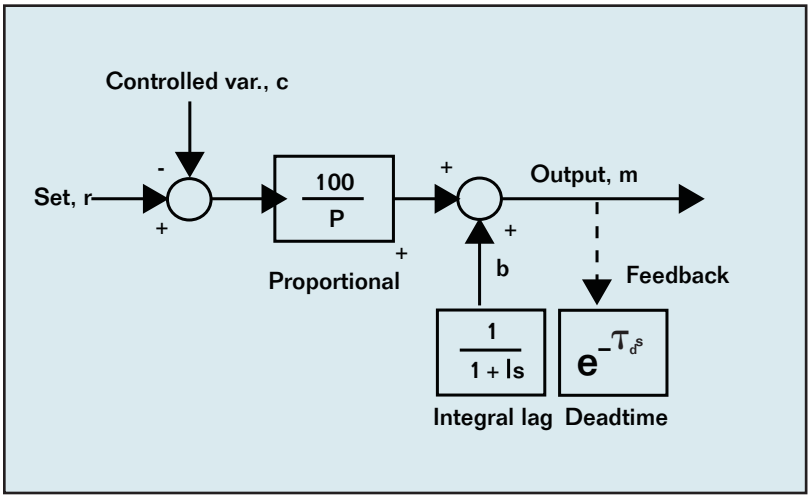
Deadtime Compensation

Access to the reset-feedback signal allows insertion of deadtime compensation τ_d as shown in Figure 5. Delaying integration in this way improves the controller's performance, giving it the capability of a Smith predictor. The same behavior can't be attained using other means of integration. However, the controller isn't as limited in robustness as the Smith, and can be applied to all processes. The PIτ_d controller in Figure 5 has a performance close to a PID, without the derivative's sensitivity to noise. However, with the addition of derivative action, the PID τ_d controller has the highest available performance of all, both in load rejection and set-point response, and doesn't even require a batch switch for process startup.

For example, this controller's performance is shown in its simulated step-load response curve in Figure 6, compared to the curves of PI and PID controllers tuned to minimize integrated absolute error (IAE). Its integrated error is only 40% that of the PID and 14% that of the PI controller, while its proportional band is less than half of the PID and a third of the PI controller. What's even more remarkable is that the process being controlled in

FIGURE 5.

INSERTING DEADTIME

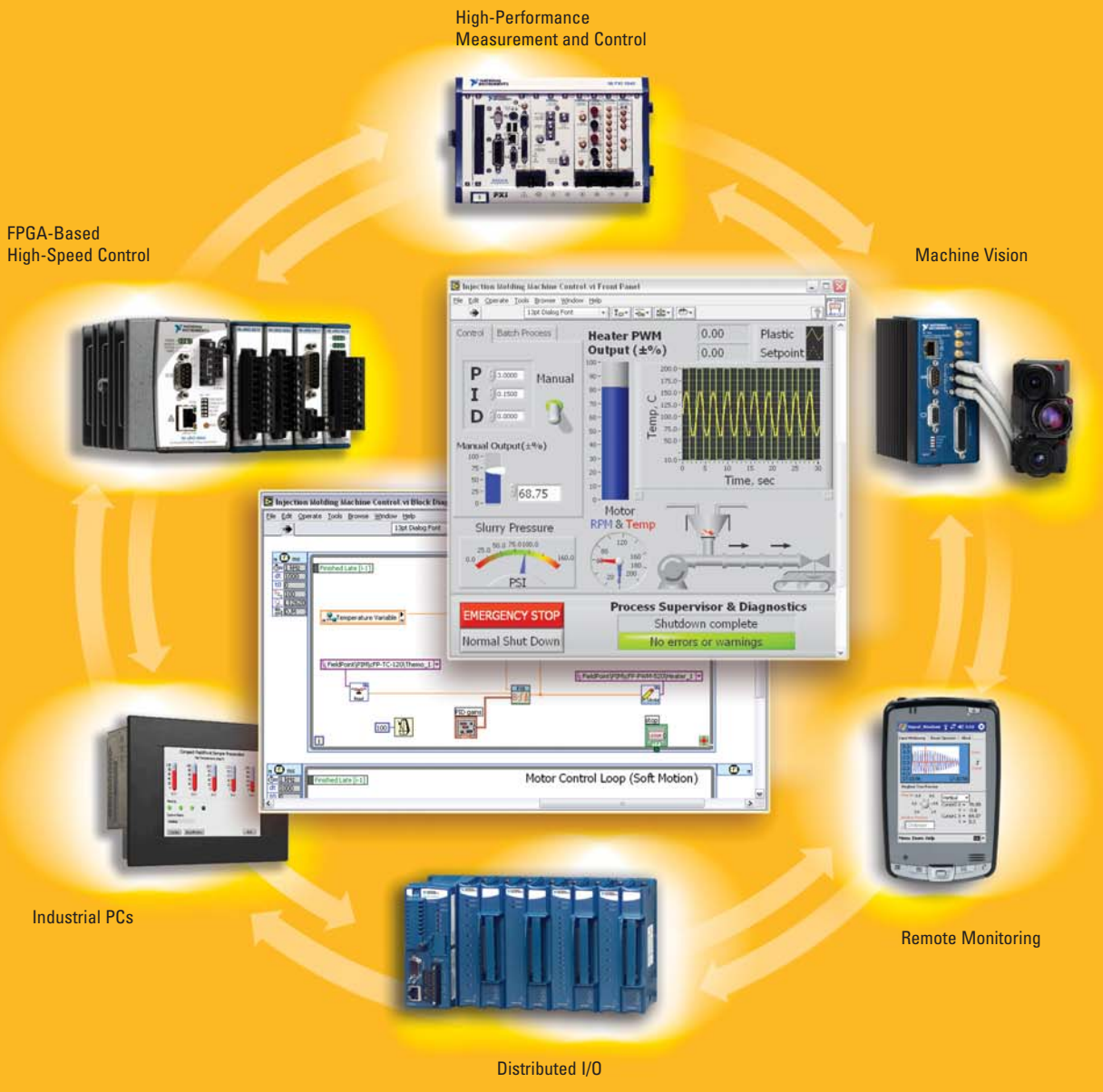


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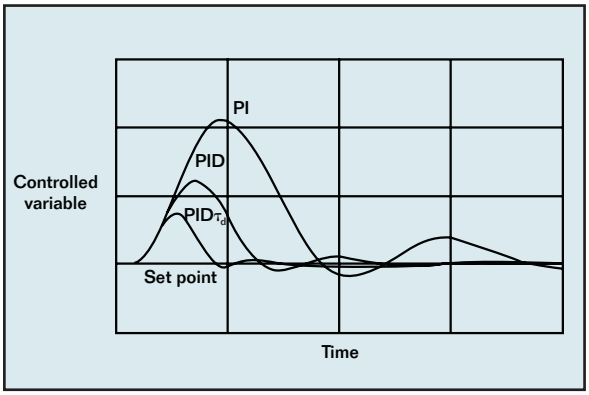
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FIGURE 6.

MINIMIZING IAE



The PIDτ_d controller outperforms others, even on a distributed process.

this simulation is a distributed lag, containing no deadtime at all! Its dynamic response is typical of heat exchangers, stirred tanks, and distillation columns, consisting of multiple lags distributed throughout their mass.

However, this performance enhancement comes with a price—low robustness. Applied to a heat exchanger, the optimally tuned PIDτ_d controller will reach its stability limit when process flow either decreases by 21%, or increases by 14%. By contrast, optimally tuned PI and PID controllers will only reach their stability limit when flow decreases by 36% and 32%, respectively. If applied to a variable-parameter process such as this one, then its tuning needs to be scheduled as a function of flow.

Cascade Control

Perhaps the most valuable role of external reset is in cascade control. The two controllers have demonstrated difficulty in transferring from full-manual to full-automatic operation. And, whenever the secondary controller is placed in manual or reaches an output limit, then the primary controller winds up.

This is solved with the configuration shown in Figure 7. The secondary controlled variable (c_2) is sent to the primary controller as external-reset feedback. If the secondary loop is then open for any reason, the primary controller stops integrating because its positive-feedback loop is open, and so it can be left in automatic all the time. Because c_2 represents the current process condition, it keeps the primary controller current and ready to resume integration whenever permitted by a closed secondary loop.

The secondary controller must have integral action, however, so that primary output (r_2) and feedback c_2 will be equal in the steady state. Any offset in the secondary loop will produce offset in the primary, as Equation 1 attests. Some cascade systems include a feed-forward calculation inserted between the controllers, such as in a multiplier (Equation 4), where:





$$r_2 = m_1 q(t)$$

and q(t) is the process measured load.

Then, the external-reset feedback path must include a back-calculation of feedback signal (f_1) using a divider, by substituting f_1 for m1 and c_2 for r_2 (Equation 5):

$$f_1 = c_2 / q(t)$$

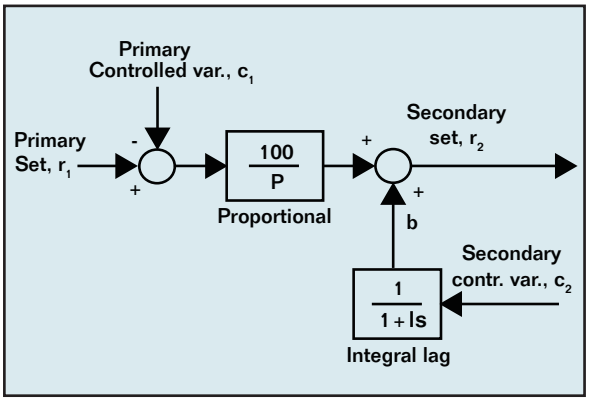
This is necessary to insure that primary output and feedback are equal in the steady state. If the selected output from the override system in Figure 4 is a set point to a secondary controller, the secondary controlled variable should be sent to all controllers as reset feedback, instead of the selected output signal.

Observe that the entire secondary loop, including the secondary part of the process, lies within the integral term of the primary controller. This turns out to be a distinct advantage. When a deadtime block was inserted in the integral feedback path, the performance of the controller improved markedly. The secondary loop now inserted in that path may include some deadtime and other lags, which also improve the performance of the primary controller. Not only does this configuration allow the integral time of the primary controller to be reduced, but also allows its proportional band to be reduced.

Adding deadtime to a controller was found to reduce its robustness, which is the ability of the control loop to remain stable while process parameters vary. However, including

FIGURE 7.

CONTROLLING BATCHES



Cascade control is improved by reset feedback of the secondary variable.

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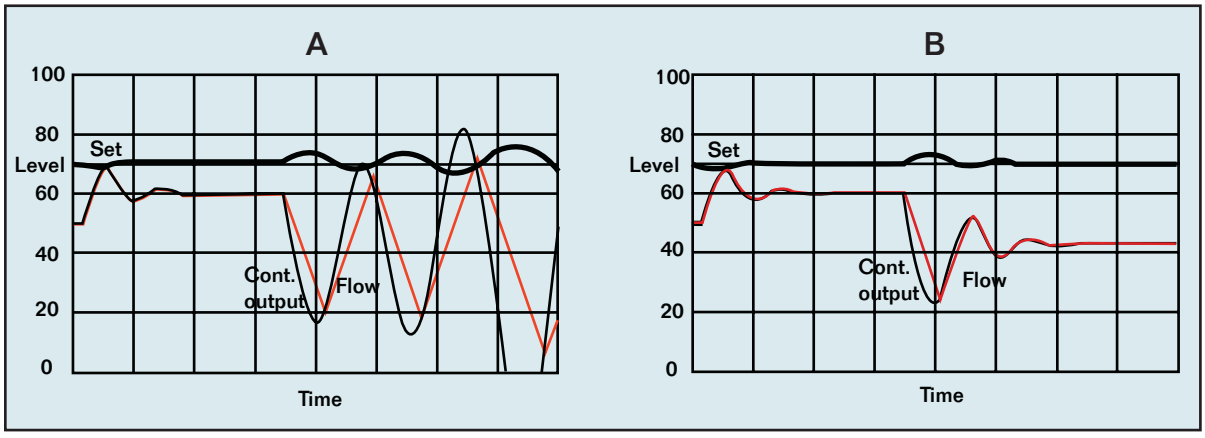
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FIGURE 8.

STOPPING EXPANDING CYCLES



Velocity limit can trigger an expanding cycle following a large disturbance, while feedback of the measured actuator position prevents the cycle from developing.

the secondary loop in the primary integral path actually improves robustness because it includes some potentially variable process parameters. The thermal time constant τ_T of a vessel is (Equation 6):

$$\tau_T = \frac{MC}{UA}$$

where M is the mass of the process, C is its heat capacity, U the overall heat-transfer coefficient, and A the heat-transfer area.

Several years ago, tests were carried out on a polymerization reactor, where batch temperature was controlled by manipulating jacket outlet temperature in cascade with the controllers connected as in Figure 7. The value of τ_T could be varied over a range of 4:1 by changing batch size M and area A. It was expected that a different set of tuning constants would be required for each combination, but the same controller settings produced acceptable results over the entire range of thermal time constants. A larger time constant meant slower heating and cooling, and slower integration as well. With this configuration, retuning wasn't required with changes in recipe or heat-transfer coefficient, both variations common to batch reactions.

Accommodating Velocity Limit

The velocity-limit or rate-limit property of a final element can pose a danger to a control loop, if the integral time of the controller is shorter that the stroking time of that element. However, the danger is hidden while the loop is operating around set point because the actuator takes little time to move only a short distance. When a sufficiently large disturbance strikes, the actuator may fall behind the controller output enough to cause more integral action and further falling behind. This triggers an expanding cycle (Figure 8A) that requires operator intervention to stop.

Figure 8A is the simulated response of a level-control loop, where the integral time of the controller is set at half of the stroking time of the valve actuator. The first disturbance is a step-load change of 10%, which isn't

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large enough to destabilize the loop. Next, a 17% step is large enough to trigger the expanding cycle. This is particularly insidious behavior because the response of the loop to normal variations in load is quite acceptable, which gives false confidence, while masking a potentially fatal instability.

The ideal correction is to speed up the actuator, but this option often isn't available. The simplest remedy is to increase integral time until the loop is stable following the largest conceivable disturbance. This may not provide complete assurance that the cycle won't recur. It also degrades the performance of the controller in response to normal low-level disturbances because integrated error varies directly with integral time.

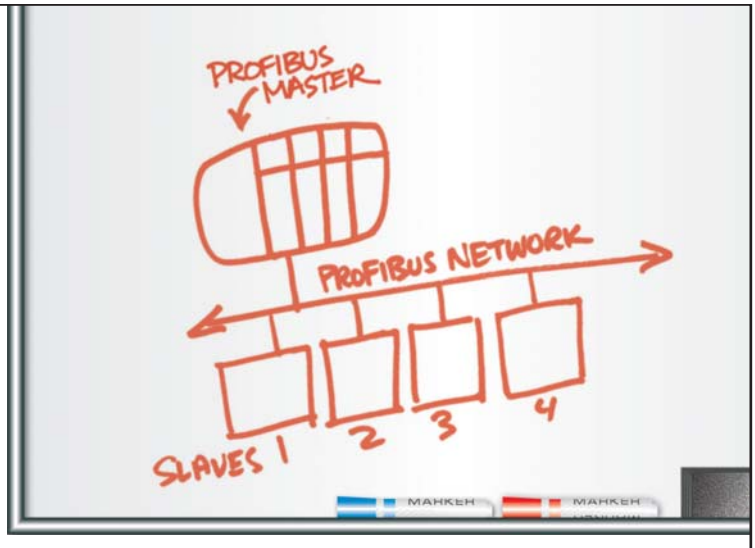
External-reset feedback of the measured position of the final element can eliminate the danger without compromising controller performance.

In Figure 8B, the actuator's actual position is fed back to the integral term of the level controller with the disturbances repeated. Integral action is now paced to the actuator's movement, which avoids any risk of an expanding cycle, no matter how large the disturbance or how slow the actuator. This is simply the same principle applied to cascade control in Figure 7. To avoid offset, there must be a secondary controller to force the final element to follow the primary output precisely in the steady state.

The need for external reset became clear during a recent startup where the level in a boiling-water reactor was controlled by manipulating the speed of up to three feedwater pumps. The integral time of the feedwater flow controller was set at 0.18 min, the expected rate limit of the pump-speed governor. When first operating with one pump on line, control was acceptable, until a disturbance triggered an expanding cycle. Review of the records indicated a much slower response of pump speed than expected, requiring an integral time in the flow controller of 1.6 min to stabilize, which was unusually high for a flow loop.

The dynamic causing the cycle disappeared when multiple pumps were on-line, so the problem is a variable one. The solution is external-reset feedback of measured pump speed to the flow controller, allowing it to integrate as fast as pump speed can follow. Implementation is complicated by the use of three pumps, with their characterizers. To avoid offset, all calculations performed in the controller output path have to be reversed in the feedback path as described earlier for feed-forward control.

F. Greg Shinskey is a process control consultant. He can be reached at shinskey@hughes.net.



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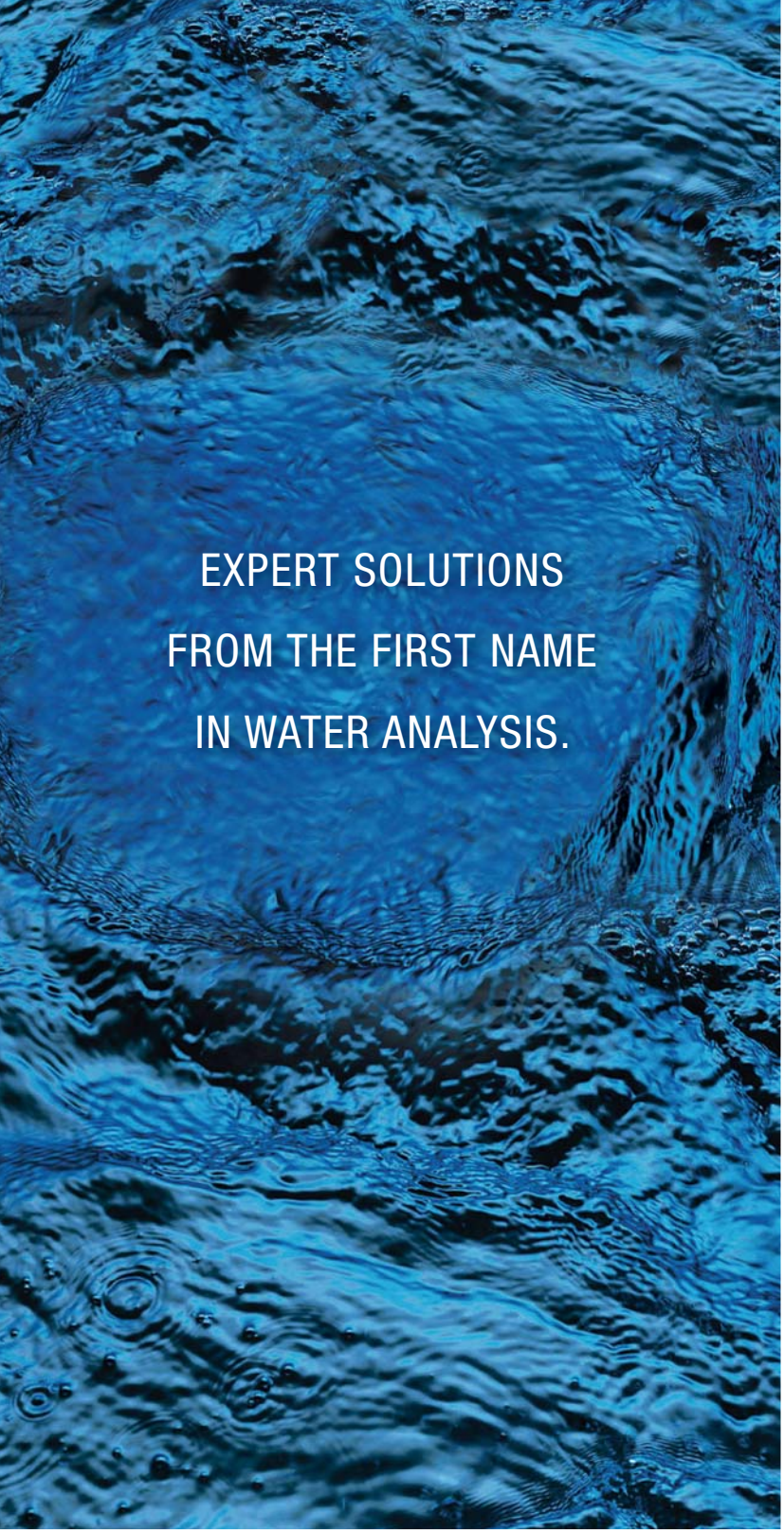
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FULL THROTTLE BATCH AND STARTUP RESPONSE

Improvements in practices for starting up a loop with a large process time constant or slow ramp time compared to the dead time can save 25% or more in batch cycle and startup time. **by Gregory K. McMillan, *Control* columnist**

The time it takes to bring a slow process variable from its minimum or maximum to set point can be a significant portion of a batch or startup phase. The more important analytical loops, pressure, and temperature loops often have a response that ramps most of the way to set point.

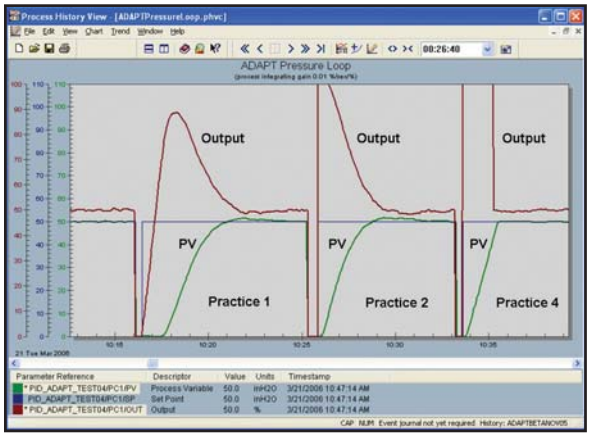
In batch processes, the ramp is most likely due to the process actually having an integrating response because the discharge valve and possibly the vent valve are closed at the start of the batch. The outlet valves also may be closed in continuous processes when a vessel is being filled or pressurized.

Even when the startup of a continuous loop is delayed until after an outlet flow is established, the small dead time and large time constant of a well-mixed volume results in a loop response that looks like an integrator. Even more problematic is the runaway response, where the process variable actually starts to accelerate. All of these responses benefit from special considerations, such as using a Lambda factor of less than 1.0 and prepositioning valves, as discussed in "Life is a Batch," *Control*, June '05, p. 80, and in a dozen "Control Talk" columns on process control techniques and tuning for batch processes.

Most of the articles and books on tuning loops concentrate on set point and load changes around operating point. For integrating processes, the focus is on surge tank level loops because these are the major source of variability in a plant because they jerk feeds around or cause a rolling action primarily from too much reset action (too little reset time). Bringing a composition, pressure, or temperature rapidly from either end of the scale to a set point is a different story.

FIGURE 1.

BATCH AND STARTUP



This graph shows the batch or startup response of a pressure loop with an integrating response for practices 1, 2, and 4.

Nothing Says Forever Like Tradition

There are four major practices for starting up a loop with a large process time constant or slow ramp time compared to the dead time. These loop practices for fast batch and startup response are:

1) Switch controller to auto with the final set point
2) Switch controller to auto with an initial set point, and then switch to final set point
3) Put controller in manual or output tracking with final set point, set valve to its normal position, wait, and switch to auto
4) Put controller in manual or output tracking with final set point, set valve to extreme position, wait, switch valve to normal position, wait, and switch to auto

Figure 1 shows the batch or startup response of a pressure loop with an integrating response for practices 1, 2, and 4. Practice 3 isn't shown because it's not viable for integrating processes. There are other practices, such as ramping the set point, for unit operations, where it's desirable that the approach of the process variable to set point and the output to its final resting value is moderated or that a profile be enforced.

In practices 1 and 2, the controller output is at its initial value at one end or the other of the output scale (often zero). All methods assume the pump and block valves have already been started and opened, respectively.

In Practice 1, if the loop is tuned to minimize variability in the controller output, which is the case for surge-volume level control, the batch phase may time out before the process reaches set point. For example, if the process time constant is 50 min, and a Lambda factor of 5 is used, then the closed-loop time constant is 250 min, and the time to reach 98% of set point is 1,000 min (or four closed-loop time constants). A similar situation exists for slow ramp rates. For integrating processes, the Lambda factor is the ratio of the closed-loop arrest time to the open-loop arrest time. The arrest time is the time it takes for a PV to make a designated allowable excursion for a given upset.

In Practice 2, the set point is changed from its initial to final value at one execution or more after the controller is switched to auto. Note that, if you switch the set point within the same execution of the module as the switch of the mode, then you'll probably end up with the same response as Practice 1. In the second batch or startup response, the set point change kicks the output, which gives the process variable a boost on its way to set point. The time to reach set point (rise time) is nearly cut in half, but the settling time is about the same. Since the overshoot is minimal, the rise time might be more important. Also, the controller tuning could be tweaked to reduce settling time.

Many astute automation engineers will preposition the controller output by what is called a "head start" or "process action." For self-regulating loops, the valve position might set at what was considered to be a normal throttle position or final resting value (FRV) seen from previous trends when the process variable had settled out at set point. This corresponds to a Lambda factor of 1 because, if held at this position, it will drive the process variable with a time constant equal to the process time constant.

For integrating and runaway responses, the process variable won't go anywhere until the valve is positioned beyond its FRV. This leads to Practice 4, in which the valve is set to an extreme position allowed by the process to give the fastest approach to set point. Next, the brakes are slammed on, so the process variable doesn't run over the set point. It's kind of like driving in Italy. The question is when do you hit those brakes?

The Wait

Short of having an Italian taxi driver operate your loop, what can be done to get the loop to its destination in the fastest way possible?

The plot for Practice 4 shows the response for a technique briefly described in "Peak Concerns," Control Talk, *Control*, April '05, p. 75. The rate of change is computed from the change in the process variable (PV) over a time large enough to get a good signal-to-noise ratio. The old value of the PV, created by passing the PV through a dead time block, is subtracted from the new PV. The Δ PV is divided by the block dead time to create a rate of change. The rate of change multiplied by the process dead time is then the predicted change in the PV that, when added to the new PV, is the predicted end point shown in Equation 1. When the end point equals or exceeds the final set point, the controller output is switched from maximum throttle to its FRV. It's held at this FRV for one process dead time, and is then released for feedback control. This method compensates for nonlinearities and disturbances that are evident when it's time to hit the brakes.

[Equation 1]

$$PV_f = [(PV_n - PV_o) / DT] * \tau_d + PV_n$$

Where:

PV_f = predicted PV one dead time into the future (%)
PV_n = new PV (%)
PV_o = old PV (output of dead time block) (%)
DT = DT block dead time (sec)
τ_d = total loop dead time (sec)

If the process dead time is underestimated, the loop will overshoot the set point. Therefore, it's important to be generous in the dead time estimate. It's especially important the dead time not be too short for zero-load integrating process (as described in the "Peak Concerns" column), where the FRV is zero and there's nothing to bring back the process variable to set point. Also, a safety margin should be added to the dead time estimate for runaway processes since the process is accelerating.

Rest in Peace

The integrating process gain can be used with the current ramp rate and the pre-positioned extreme controller output to estimate the FRV per Equation 2. Note that, if the extreme output (OUT_x) is less than the FRV, the signs of each expression are reversed to get a positive FRV. The calculation should be limited and filtered. For primary loops in a cascade control system, the extreme output must not exceed the set point limits of the secondary loop. It's necessary to keep the units of the process variable and output and the process integrating gain consistent. If the process integrating gain is in %/sec/%, the process variable and output must both be in %. For integrating processes and $OUT_x > FRV$:

[Equation 2]

$$FRV = OUT_x - [(PV_n - PV_o) / DT] / K_i$$

Where:

FRV = final resting value (%)
K_i = integrating process gain (%/sec/%)
OUT_x = output at extreme allowed by process (%)
PV_n = new PV (%)
PV_o = old PV (output of dead time block) (%)
DT = DT block dead time (sec)

Next Destination

The optimal switching technique based on the rate of change of the PV is ideally suited for an integrating or ramping process, but works well for self-regulating and runaway processes where the fastest possible approach to set point is desired. It also reduces the dependency on tuning since the controller only has to correct for errors in the dead time and the FRV. These and other process control techniques can save 25% in batch cycle and startup time and get you to the next destination faster, which might just be happy hour.

Greg McMillan is an inductee of the Process Automation Hall of Fame and is retired from Monsanto. He is a prolific author on process automation He can be reached at gkmcmi@austin.rr.com.

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MACHINES STAYING HEALTHY BY STAYING SMART

Machine health monitoring is going beyond mainstream vibration and oil analysis to embrace increasingly high-resolution sensing and prioritized data processing technologies to help users make more intelligent, efficient, cost-effective decisions.

by Jim Montague, executive editor

Recent advances in sensing technologies are allowing users to look ever deeper and with greater resolution into their machines and process applications. This and focused handling of the resulting data is allowing many users to find and diagnose potential problems even sooner, and make more intelligent decisions on scheduling maintenance, allocating resources, managing inventory, and scheduling to minimize the cost of maintenance and repairs.

Organizationally, it's true that machine health monitoring and/or management fits within larger, less tangible concepts, such as condition monitoring, preventive and proactive maintenance, asset and lifecycle management, and even enterprise resource planning. However, machine health comes in far more direct contact with the processes it monitors than any ostensibly higher-level methods, and often helps supply them with data they need to function.

Healthy Evolution

"As users move from straight, time-based or run-to-failure maintenance, they find they need information on the condition of their assets. Next, it's important to integrate that physical, process data into a condition monitoring platform, which can correlate it, and help measure performance," says Scott Breeding, product line leader for Bently Nevada (www.bently.com), a division of GE Energy. "This is what gives users an indication that a bearing may be running hot before they can see any signs of wear or damage. For example, an engineer may use process data indicating a vibration to deduce that a centrifugal pump is cavitating because an operator is applying inadequate suction-head pressure, and this can support a decision to improve that process.

"Process data also can be used to write new rules around assets, derive subsequent indicators, and screen data to produce alerts when a problem may be likely to occur. This allows users to focus more closely on the 'bad actors' in their applications."

Similarly, using screened process data, and performing failure mode of effects analyses (FMEA) can help users pick the most appropriate machine health technology for each asset based on its relative criticality. These typically include trebology (lube oil analysis), vibration analysis, thermography, online and offline motor current monitoring, and several other less well known technologies.

For example, Bently Nevada reportedly has solved 10 separate machinery problems on almost 50 process machines at UPM-Kymmene's Wisaforest pulp and kraft/sack paper mill in Pietarsaari, Finland, and. This project also was part of the mill's overall installation of a new recovery line to maintain its production of 800,000 air-dried tons per year.

Though most of its machines use rolling-element bearings monitored by accelerometers, the mill's huge lime kiln uses fluid-film bearings monitored by X-Y proximity probes and accelerometer transducers. To let operators see basic condition information data and let rotating machinery engineers see diagnostic data, the mill's process control system set up a bi-directional OPC interface, which imports amplitude and alarms from System 1 into the DCS, and exports process variables from the DCS into the System 1 database. The main problems that Bently Nevada helped solve at the mill included:

- Identifying intermittent, high-frequency vibration amplitudes in the inboard bearing in the recovery boiler's 600-kW air fan, which helped the mill's engineers better schedule lubrication system repairs and bearing replacements.
- Correcting elevated vibration levels in the 15.4 x 443-ft lime kiln's drive rollers by evaluating their phase, rpm,

FIGURE 1.

BAD VIBES



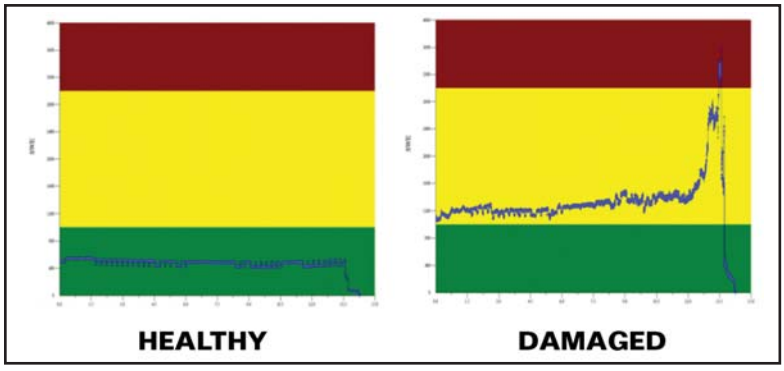
An amplitude/phase/time (APHT) plot of vibration data from the drive rollers on this Wisaforest mill's lime kiln demonstrated a structural resonance as it sped up, which required stiffening and strengthening the drives' supports.

amplitude and frequency data, which found a structural resonance at 36 Hz as the kiln moved from low- to higher-speed operation, and required stiffening and strengthening the drives' supports (Figure 1).

- Locating a cracked inner ring in a bearing in the recovery boiler's exhaust gas fan, shown by a characteristic "ringing" phenomenon in the amplitude/phase/time (APHT) plot for the fan, which was repaired.
- Identifying excess vibration and a faulty rubber element in a coupling on the recausticizing unit's white liquor pump, which also was repaired.
- Finding inboard bearing deterioration in the recausticizing unit's totally unmonitored mixer, which was adjacent to its monitored rotary filter, and required repairing the mixer's a broken lubrication pipe.

FIGURE 2.

SOUNDING STRESSED



Ultrasonic stress wave energy (SWE) plots for two machines show one was damaged by using the wrong lubricating oil. SWE levels on the "faulty" machine were immediately higher than its twin, and accelerated as actual damage occurred.

Implementing these solutions reportedly helped the Wisaforest mill achieve payback on its upgrade investment in just eight weeks.

In addition, though basic vibration has been used to check machine health for decades, new algorithms and software are helping users monitor more operating parameters and applications. For example, proximity sensors and software included in Bently Nevada's Asset Condition Monitoring Solution can take signals that indicate the trueness of shafts in motors. "We have new algorithms and software that can set up alerts and alarms based on asset conditions," says Jeff Schnitzer, Bently Nevada's general manager. "We also have Rule Packs that allow users to write rules based on vibration, temperature, or other indicators, and create subsequent functions and responses.

"These and other machine health monitoring capabilities can then be combined with maintenance records, assets classifications, and other data to create baselines for entire plants, and help target future investment where it will return the most improvement. This means no longer touching or replacing components based on past outages and history, but touching or replacing them based on real-time reliability data."

Sharpening Senses

Robert Skeirik, machinery health product manager for Emerson Process Management (www.emersonprocess.com), says it links the five main senses of machine health—vibration, oil condition, motor current, infrared thermography, and ultrasonics—into its AMS Machinery Health Manager, AMS Suite software, CSI 2130 portable analyzer, CSI 4500 analyzer that connects to a multiplexer, and CSI 9210 machinery health transmitter.

Skeirik adds that Emerson and its CSI division recently helped upgrade the 20% worst-performing of 27 coal pulverizers at American Electric Power's (AEP) John E. Amos plant in St. Albans, W.V. The units were previously serviced on a 12-month schedule, and collectively experienced 3-4 unexpected failures per year. Besides preventing $240,000 worth of breakdowns in the project's first year, CSI and Emerson helped AEP and Amos' engineers achieve savings and return on investment (ROI) in several machinery health monitoring areas:

- Lube oil analysis saved more than $74,039 for a 4-to-1 ROI
- Infrared thermography saved more than $343,000 for a 5-to-1 ROI
- Motor current analysis saved more than $344,000 for a 20-to-1 ROI
- Motor testing saved more than $381,000 for a 20-to-1 ROI
- Leak detection saved more than $1.2 million for a 60-to-1 ROI
- Vibration analysis saved more than $4.2 million for a 42-to-1 ROI
- Total savings were $6.5 million for a 27-to-1 ROI

"Machinery health is now just one component of a plant's overall status, which is continually seeking to answer the plant manager's two basic questions about reliability and availability," says Skeirik.

An Ultrasonic Heartbeat

Though vibration detection and analysis is a bread-and-butter machine health monitoring method, and one that's also increasing in sensitivity and sophistication, some developers believe ultrasonic monitoring may be even more useful. Based on military and aerospace monitoring methods, Stress Wave Analysis (SWAN) technology from Swantech (www.swantech.com) reportedly can detect even earlier when a machine is starting to demonstrate symptoms that will lead to failure (Figure 2).

"Ultrasonic technology provides the earliest detection of machine problems by detecting the unique sounds made by friction, impact events, and minor surface damage," says Ralph Genesi, Swantech's president and CEO. "Traditional machine health has been based on vibration, but this is too late because some damage may already have occurred and a safety shutdown may be needed. Ultrasonic detection gives users more time to gain the knowledge to plan maintenance and a response, and lets users decided if they need to shut down now or if they can safely run their planned quota.

"In addition, electrical signal histories collected over time, or histograms, can show if new signals are skewed, and even help indicate if a machine has lube problems, cracks, or seal damage. Ultrasonics can even show when an external load is placed on a machine, which can help operators change bad habits, and extend the lives of their devices."

Bearing Protection

For example, to help it process more than 2 million tons of reclaimed steel per year, North Star BlueScope Steel in Delta, Ohio, recently asked Swantech to help maintain its flat roll machines, which usually run 24/7 and can cost $1,000 per minute of unplanned downtime. A day's downtime is very expensive. The steel maker's SWAN system gathers information from Stress Wave data collectors associated with each bearing, analyzes it on a SWANserver in the plant's offices, reports on any early signs of wear in the mill's equipment, and allows managers to plan maintenance or repairs up to six months in advance.

"We're concerned mainly about failed bearings and failed gears," says Matt Morris, North Star BlueScope's reliability-team leader. "All of the gearing is unique to the machine, and we have no spares. Because getting a new one usually takes at least six months, limping along or shutting down completely can add up to large losses. However, the field of available methods is very narrow because many of our large bearings turn very slowly. Also, the sensors mounted on each bearing have to withstand the high temperatures and heavy vibration in our processes.

"Initially, the SWAN system was standalone, but it was eventually added to our Level 1 network for backup purposes. Now, it's also accessible through routers from our main Level 3 business systems."

Not only has access to this information reduced production losses, but it also has allowed NorthStar to shrink its $3 million inventory of spare parts, and avoid paying thousands of dollars for rush orders.



AT THE CROSSROADS: PROCESS AUTOMATION AT DOW, PART 2

Dow Chemical's Transition to a New Development Approach

by Margaret Walker, Ed Sederlund, Jerry Gipson, and Eric Cosman, Dow Chemical Co.

More than 30 years ago, the Dow Chemical Co. embarked on what ultimately evolved into a global corporate initiative: to develop and broadly leverage a standardized, highly integrated process automation system, incorporating basic process control, process information, and safety-system functionality. Based on what we wanted to accomplish, and a lack of commercial offerings at that time, we developed several proprietary solutions, including a process control system that came to be known as MOD, which is short for "Manufacturing Operating Discipline." With the MOD 5 System, Dow was the first company to certify a process automation system with logical separation of the control and safety functions. Our development approach served us well for many years, but we reached a point where it was no longer cost-effective to develop our own system. Part 2 of this three-part series describes Dow's transition from our successful, homegrown solution to a commercially developed solution, including what went into the decision process, why we needed to change, and how we grew and nurtured a truly collaborative, technical relationship that will take our operating discipline forward.

FIGURE 1.

MOD 5



With its MOD 5 System, Dow was the first company to certify a process automation system with logical separation of the control and safety functions.

From the beginning, continuous improvement has been at the heart of the Dow Chemical Co., and forms the essence of its mission today, "To constantly improve what's essential to human progress by mastering science and technology." To achieve this, we're committed to the principles of Sustainable Development. As we discussed in Part 1 ("The Mod Squad: Process Automaton at Dow," *Control*, Feb. '06, p. 68), this underlying philosophy has been an instrumental driver in our process control journey.

The roots of Dow's process-automation philosophy go back to the 1960s, and are based on our operational strategy. Consistency and sustainability are key elements of that strategy. As a global company, it's important to operate facilities the same way, every time, to achieve consistent high quality, as well as process reliability and repeatability of best practices.

Of course, safety performance also is a key focus for Dow. Protection of people, the community, and the environment have always been non-negotiable priorities in our corporate culture. We must operate our plants safely every time, as well as the same way every time. Our "Vision of Zero" translates to zero incidents, injuries, illnesses, accidents, and zero environmental harm. So our safety practices needed to be repeatable and re-deployable, too. All of these elements make up our Operating Discipline (OD).

Initially, commercially available process automation systems didn't give us the capability we needed. Technology was available that could be applied to the problem, and we developed several solutions based on general-purpose computers and operating systems. As we gained experience with these systems, it became evident there were significant shortcomings when applied to pro-

cess automation and customer needs. Over time, we developed the MOD 5 system to meet our needs.

For nearly 30 years, and with more than 15 patents related to MOD technology and TUV Safety Integrity Level (SIL) 3 certification, we enjoyed continued success in fulfilling our OD with our proprietary MOD System series. It grew from an analog system at one site in the 1960s to a globally deployed, standard process control system with 1,500 systems installed throughout Dow by the year 2000. Over the years, the MOD series system delivered tremendous productivity benefits to Dow. We reduced the time to develop control strategies by 45%, and reduced support costs by 50%. Process startups were automated and coordinated across multiple-unit operations. Alarms were "intelligent," reflecting the state of a process, rather than just a deviation from a limit. Batch processes were fully automated, resulting in reduced cycle times. Imbedding the OD concept in the automation system also helped us achieve our Environmental, Health and Safety (EH&S) goals. All of these performance improvements jointly contributed hundreds of millions of dollars in value to our businesses.

It seemed that we'd come to the crossroads, where commercially available technologies were at long last catching up to us.

Exploring New Sustainability Solutions

In early 2000, however, we realized it wouldn't be cost-effective to continue to invest in proprietary hardware and software systems. Our efforts to develop the next version of the MOD system, MOD 6, proved to be much more time and cost intensive than we anticipated. MOD 6 development began in 1987, and envisioned a triple-redundant, synchronous architecture built on dual-redundant synchronous architecture learnings from MOD 5.

The strategic error in MOD 6's development was building our own computer and communications application-specific integrated circuits (ASICs) and related software and development tools. MOD 6 was successfully demonstrated in 1998. However, rapidly changing computing and automation technologies, such as changes in networking, connectivity, communication protocols, integrated systems, real-time information access, and embedded intelligence made it difficult for us to incorporate all of them into the current development cycle, which became more unwieldy to manage with each passing day. It seemed that we'd come to the crossroads where commercially available technologies were at long last catching up to us.

External markets also conspired to force this change. As a global manufacturer, our new plants were bigger and more complex than ever. We also were expanding rapidly. The expectations for capital utilization, energy use, overall efficiency, and throughput were higher. And, the need for information integration between processes, plants and people were growing more sophisticated. The performance pressures in the global economy were intense, and, just like everyone else, we needed to find ways to do more with less. The need to continuously improve productivity was always there—every cent matters to the bottom line. So, it was imperative that our plants run smoothly, efficiently, and without interruption. At the same time, we had to accomplish all of this while being true to our OD, and make sure that our solution was sustainable going forward.

Consequently, it just no longer made business sense to continue to develop and maintain our own process automation system. We needed to focus on our core manufacturing business. While we wanted to preserve our process automation expertise, we knew we didn't want to be providers of hardware and software, so we began looking at other options. Some of the commercially available solutions looked like they had the potential to develop into something that could work for us, but going this route first required a lot of investigation, homework, and soul-searching. It isn't easy to give up a job that you've been doing very well for more than 30 years, and just hand it over to someone you don't know. Yet we knew, for pragmatic reasons, it had to be done.

The culture, the knowledge, and the passion for process safety were already ours. That wasn't going to change in this transition. What was going to change was the platform. It just so happened that our first platform was the system we invented. So, the goal was to slide that platform out, slide in another platform, and allow the culture to continue forward undisturbed. We needed to find a solution platform to meet our needs, as well as a process automation partner that would share our vision, goals, and philosophy.

Critical Requirements: "The Crown Jewels"

To be considered as a possible solution, a commercial system had to align with our process automation wants as well as our needs. Once we decided to pursue a commercial option, we defined more than 400 requirements for this system that addressed our need for sustainability going forward. Long-term commercial availability, cutting-edge technology, and forward-looking solutions were all essential criteria. We absolutely needed a process control system that would take us successfully into the future, be deployable on a global scale, leverage commercial standards as they became available, and be on a platform that Dow could use as a standard at any plant anywhere.

We expect our plant assets to run for 40-50 years, so our process control systems need to be as sustainable as the rest of these assets. A commercial system also would have to meet our increasing need for knowledge management, while remaining true to our process engineering culture. It

had to be able to shape our future engineering culture, just as the MOD System had since the 1960s.

After agreeing to go commercial, we explored various ways to collaborate with external suppliers. Following our initial attempts, we hired an objective, third-party consultant to work with us in the evaluation and selection process, and help us define our requirements. Of course, after all our team had achieved with the MOD series, our expectations were much higher and our wish list far more detailed than a typical automation customer. We knew exactly what we wanted. Were what you might call an extremely knowledgeable consumer.

Based on our list of 400 requirements, Dow defined 32 high level criteria, affectionately known as "The Crown Jewels." The consultant provided a short list of recommended candidates for Dow to approach. We did an extensive onsite evaluation based on our requirements for each recommended company, and met with their executive management, technology officers, and development teams.

"It was unique to meet a customer with such a detailed list of functions outside the normal market standard requirements. This led to a number of internal discussions, including how to fulfill these functions and the technical needs for all of them," recalls Frank Duggan, currently senior vice president for ABB's Group Account Management. "On the other hand, we were dealing with a partner with a deep understanding of systems. If we could fulfill the rest of Dow's requirements, we felt this could become our competitive differentiator for the industry."

FIGURE 2.

THE "CROWN JEWELS"



Based on its list of 400 requirements, Dow defined 32 high level criteria, affectionately known as "The Crown Jewels."

ABB: A Shared Vision

ABB was one of the companies on our short list. We spent five days with them during the onsite evaluation. One thing we told them upfront was that we didn't want to see any "smokescreens" or ethereal vision presentations. While vision is extremely important to us, we also had an immediate need for a sustainable solution. We couldn't afford to wait for something that might materialize someday.

Consequently, we met, we visited, and we sat through numerous company overview and technical presentations, some quite boring to be perfectly honest. On the fifth day of their visit, ABB's presenters showed us their Industrial^{IT} technology, which was being developed as the heart of their Extended Automation System 800xA. At that point, we knew that we'd found the commercial solution that would take us into the future. We saw in that one day's presentation exactly what we were seeking—the integrated environment, safety systems, and repeatable engineering solutions. It was all there. Industrial^{IT} had the ability to integrate multiple systems and plants into one environment, as well as system flexibility, integration of databases, common operating views, and engineering functionality. It was right in line with what we needed. The alignment with their direction compared to where we wanted to go also came together nicely.

So, we'd found the commercial technology that could accomplish what we needed to continue fulfilling the Crown Jewels. We could use this platform to leverage our experiences and learnings with the MOD System's services. However, while the technology was a crucial ingredient to making the conversation happen and implementing the solution, much more was needed to help Dow and ABB's relationship succeed.

Consequently, even though we'd found the technology we thought could take our OD forward, the really hard work was just beginning. We needed to lay the foundation for a truly collaborative relationship, and let go of our own sole-system development mindset at the same time. The next article, Part 3, will discuss what was involved in building a close working relationship with ABB, and the elements that are essential to starting and sustaining any collaborative relationship that yields results.

For more information on Dow's patents related to the MOD technology and TUV SIL 3 certification, go to http://www.uspto.gov/patft/index.html.

Margaret R. Walker is vice president of engineering solutions for Dow's Technology Centers and Manufacturing and Engineering Work Process; Jerry N. Gipson is director of Dow's Engineering Solutions Technology Center; Edward R. Sederlund is Dow's Process Automation product manager; and Eric C. Cosman is Dow's Engineering Solutions architect. The authors would like to thank Peter J. Kindt, Dow's Process Control and Advanced Control discipline leader, who also contributed to this article.

PC-Based Control Infiltrates Process Plants

PC-based control has been slow to catch on in the process industries. Many process end users have no qualms about using Intel-based controllers, but shy away from trusting control of critical processes to Windows, Mac, or Linux operating systems. In an ironic twist, however, performance and features specified by process plants are forcing their machine and process-skid suppliers to use PC-based controls.

End users buy process skids for water treatment, air handling, and specialized chemical processes such as fermentation. Process plants often need their machines and process skids to have a sophisticated local operator interface, connectivity to the enterprise, and remote monitoring capabilities. High throughput and extreme precision are typical requirements for these machines. Process skids may need sophisticated, advanced control algorithms to fulfill specifications.



Once you become comfortable with PC-based control, it's a short step to PC-based control of your own critical processes.

To meet these requirements, many machine and process skid builders are turning to PC-based control. Initially, many builders used PCs to meet those customer demands, but would typically use a PLC for real-time control because they didn't trust PCs for critical control.

This trend is rapidly changing because of lower costs, increased reliability, and increasing ease of use. Many of these factors were demonstrated by PC-based control suppliers at the recent National Manufacturing Week 2006 tradeshow near Chicago.

It's much cheaper to combine operator interface and control into one PC-based system than using a PC and a PLC. Not only are purchased costs lower, but integration between the PC and the PLC is eliminated. In addition, PCs can outperform similarly priced PLCs.

Axiomtek (www.axiomtek.com) and National Instruments (www.ni.com) displayed panel PCs at NMW with 6.4-in. LCD screens. A panel PC looks like an operator interface panel, but it performs like the PC that it is, and a "graphics panel" gains real-time control capabilities. And, because the graphics panel has a PC-based engine, off-the-shelf HMI software packages optimized for limited resource targets can be used. InduSoft (www.indusoft.com) was demonstrating this type of software at NMW.

Of course, a control system also needs I/O, a reliable real-time operating system, and an easy way to program discrete and process control applications.

Open standards aren't perfect, but they've had a big positive impact on I/O for PC-based controllers. These typically are supplied with an Ethernet port. Just make sure to match the controller and the I/O Ethernet protocols, because Ethernet by itself is not a complete standard.

Similarly, the operating system has long been a bane of PC-based control. Microsoft has addressed many of these concerns with its embedded CE and XP operating systems. Linux is also a popular choice. For ultimate reliability, hard, real-time operating systems are available from vendors such as Green Hills Software (www.ghs.com) and TenAsys (www.tenasys.com). All of these operating systems are real-time and reliable enough for most process control applications, and they aren't memory hogs. Because memory requirements are reduced, RAM costs are cut, and flash disks can be used instead of less reliable hard disks.

The last hurdle for many is programming the PC to perform real-time control. C++ and other high-level computing languages used to be the only option, but the rise of Visual Basic has given many programmers another and often better option.

In addition, many suppliers now provide graphical programming software conforming to the IEC 61131-3 standard (www.plcopen.org). These software packages allow programming of PC-based controllers with a standard methodology. IEC 61131-3 includes the Sequential Function Chart language used to structure the internal organization of a program, and four interoperable programming languages: Instruction List, Ladder Diagram, Function Block Diagram, and Structured Text.

As a result, PC-based control is becoming more widespread in process plants, often via purchased machines and process skids. Once automation professionals become comfortable with PC-based control through purchased systems, it's a short step to implementing PC-based control for control of their own critical processes.

Dan Hebert, PE, senior technical editor
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PLCs and HMI Platforms? Solenoid Testing?

"Ask the Experts" at ControlGlobal.com is moderated by noted process control authority Béla Lipták. He and his experts in process control and optimization, recruited from the co-authors of the 4th edition of the three-volume *Instrument Engineer's Handbook*, answer process control questions from all comers, and the best are published here.

Question:

I would like to ask for your comments and suggestions. I have volumes 1-3 of your *Instrument Engineer's Handbook*, and find them to be an excellent source of technical information.

We're constructing a large wallboard manufacturing plant in West Virginia. We're currently at the crossroads of deciding whether we go with Siemens' S7 PLC and WinCC HMI platform, or with Rockwell Automation's Allen-Bradley ControlLogix PLC and Wonderware's InTouch HMI platform for the plant automation and controls. We're presently collecting data, and analyzing the pros and cons of both systems. Some areas of special concern are: automation performance comparison; software friendliness and ease of use; local plant technical support availability; availability of personnel resources familiar with either system; networking superiority; automation superiority; price comparison; etc.

Any specific comments/suggestions you may have would be highly valued for someone trying to decide between the two systems. Also, if you know of anyone in your network who been down this decision path, please forward this inquiry because I would like to ask for their comments/suggestions as well. Thank you for your time.

Sam Roe, senior electrical/automation engineer
BPB America
Sam.roe@bpb-na.com

Answer:

The architecture of the system is important. This applies to hardware architecture as well as software architecture.

The central component of the hardware architecture is the control-network. Without a doubt, this should be based on standard Ethernet. I stress standard Ethernet, not modified "industrial" Ethernet, which requires specialized hardware, and which doesn't use IP. For availability, it's important that true, duplicated "DCS style" redundancy is supported, not just ring topology. The application protocol on top of IP is critical for interoperability between devices from different manufacturers, as well as package unit integration.

Do not go for proprietary protocols over Ethernet and IP. These criteria pretty much narrow the control-network platform down to Foundation fieldbus' High-Speed Ethernet (HSE). Siemens doesn't support HSE. Allen-Bradley does to some extent, but there are other options you should examine.

At the core of the software architecture, you need a technology to exchange data between hardware and software, and between the different software applications. Undoubtedly, this should be OPC, including the principal flavors OPC-DA, OPC-A&E, and OPC-HDA. Both Wondwerware and WinCC support at least OPC-DA. Wonderware maps it to its own solution, loosing some ease of use, though this situation could have changed. I have no experience with WinCC.

In addition, there are other options you should explore, and other software technologies you should make sure are supported. These include: SQL/ODBC/OLE_DB/ADO database access; ActiveX graphics components and OLE container; DDE for legacy; VBA for script; and HTML for web viewing, etc.

For OPC and other software technologies take a look at the book, *Software for Automation: Architecture, Integration, and Security*, which is available at www.isa.org/autosoftware.

For Ethernet, check out *Fieldbuses for Process Control: Engineering, Operation, and Maintenance*, which is available at www.isa.org/fieldbuses.

Jonas Berge
SMAR
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Answer:

Is there some reason you've limited your evaluation to the two combinations indicated? A good reason might be that there's already experience with all of these products elsewhere in your company. On the other hand, if there's not a particularly strong reason to impose such a limitation, you might want to include at least a couple other possibilities:

- In the PLC area, Modicon (Schneider Electric) especially, and perhaps GE Fanuc as well.
- In the HMI area, Intellution's iFix (now owned by GE), and RSView32 (Rockwell Automation/Allen-Bradley), and perhaps CiTect, too.

As for PLC's, I have personal experience with Modicon (584, 984, Quantum) and Allen-Bradley (PLC-5, SLC-500, ControlLogix) in pulp and paper and food industry applications. I've found both to be solid products, but have I a slight preference for Modicon.

As for HMI's, I have personal experience with Wonderware and Intellution, more hands-on experience with Wonderware, and a slight preference for Intellution [Editor's note: now GE's Proficy] for technical and commercial reasons.

This may not have helped because it's more opinion based on personal experience. My Modicon preference comes from early days employed in the Cellulose and Specialties division of Procter and Gamble, where we chose Modicon over Allen-Bradley as the standard for that division of the company because it could better meet higher performance requirements in the pulp and paper mills.

Of course, keep in mind that most leading HMIs will integrate well with most leading PLCs. There may be some minor advantages in pairing ControlLogix and RSView because both are Rockwell/A-B products.

R. H. (Rick) Meeker, Jr., P.E.
Reliable Power and Controls Corp./Process Control Solutions, Inc.
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Question:

In reference to our phone call few days ago, I would like to seek your assistance with the following:

1) I want to perform a life assessment calculation for a solenoid valve to identify how many years it has left. This calculation should be based on the environmental conditions in which this solenoid is kept. For example, real outside temperatures reach 55 °C in summer with 100% humidity.
2) I was advised to search Google for "Reliability, Maintainability and Risk" by David J. Smith and for "Weibull Analysis," but I still can't seem to narrow the search to find a typical calculation for any general instruments such as control valves. Can you help focus my search?

Mufeed Al-Ghumgham
Mufeed@Sahara.com.sa

Answer:

I first addressed this question in about 1980. My conclusion is that there is no way to accurately calculate or predict solenoid valve failure in the general sense.

Most failures I saw during my 20 years in the plants were electrical. Rare failures have been seen with dirt blocking either flow or, more likely, switching. Dirt and water are enemies of reliable operation of any pneumatic systems.

Another sort of failure was a failure to switch because the DC voltage applied was not quite sufficient to overcome the magnetic structure in the valve. Note that the required pull-in voltage is higher than the hold-in voltage. These different values might not be listed in the catalog specifications. Electrical failures may be linked to voltage spikes, or the reverse—low voltage heating the coils, but insufficient to cause armature motion. Exposure to voltage/current spikes is heavily dependent on installation practices and even local-earth conductivity. Voltage-suppressing diodes also are sometimes used. These may raise questions of their possible failure due to a shorted condition. Long wire runs will increase exposure to lightning induced spikes. It's generally believed that heat contributes to coil failure. One approach supported by SP84 is that proof of reliability can be achieved after extended experience in exactly the same service. Proper record keeping is required for this, but is not often seen.

The only recommendations I could make is to select valves from a proven vendor, well within their ratings, and reduce exposure to voltage spikes and excessive heat. If very high reliability is required, then redundancy in the system design is required to reduce the impact of any single failure.

Cullen Langford, PE

Contact-Type Level Instrumentation Rebounds

Radar and ultrasonic sensors no longer dominate new product offerings.

For the past few years, *Control*'s annual level-instrumentation roundup has observed many non-contact radar and ultrasonic-level instruments, and a scarcity of new versions of traditional instruments. Market researchers were probably gloating (or surprised!) to note that one of their predictions had actually come true: that radar and ultrasonic sensors would be the fastest-growing segments.

For instance, Venture Development Corp. (VDC, www.vdc-corp.com) recently reported in "Rising Levels of Process Level Measurement Devices," *Control*, Nov. '05, p. 61, that, "The largest relative gains in continuous measurement technologies are expected for microwave/radar contact/guided (10.6%) and non-contact sensors (8.4%)."

Though true before, this trend isn't reflected in the products in this year's roundup. Now, there are more non-non-contact sensors in our roundup than ultrasonic and radar sensors. And, despite contributing author David Spitzer's glowing coverage of laser sensors in "Zap! How Do It Know?" *Control*, Feb. '06, p. 57, we see no new laser-based sensors in this roundup.

In fact, because not all of the non-contact sensors listed below are actually new—some are enhancements to existing products—this year's roundup may indicate a return to relative normalcy in the level industry.

This year's roundup shows more agreement with VDC's forecast that, "While there likely will be a barely perceptible shift from mechanical to electronic sensing technologies in continuous measurement, shipments of mechanical point measuring/sensing devices will outstrip electronic types." VDC's global study, "Worldwide Process Level Measurement and Inventory Tank Gauging Markets, October 2005," also anticipates above average growth for float switch (6.7%), vibration (6.4%), and paddlewheel (6.1%) measurement devices." We're again seeing some of the devices below for the first time in a coon's age.

You'll see a mixed collection of all sorts of level sensors in the following roundup, including radar, ultrasonic, float, magnetostrictive, weight-based, vibrating fork, thermal and pneumatic devices. That's the way it should be. Instead of rolling over and playing dead in the face of increasing non-contact sensor sales, other vendors see a growing economy, have more confidence in the market, and are starting to roll out new, contact-type sensors. Hooray!

—Rich Merritt, senior technical editor

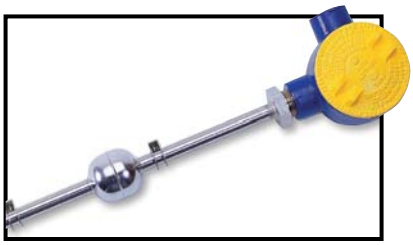
LEVEL TRANSMITTER SOUNDS GOOD

Type 8175 ultrasonic level transmitter has 0.25% FS accuracy over a 1.0 to 32.0-ft range, with resolution of 1/8 in. This non-contact transmitter is suitable for continuous level control, on/off level measurements of fluids/solids, flow measurements in open channels, and volume and distance measurements. It includes an ultrasonic sensor, transducer and eight-digit, multi-language display protected by a NEMA 4 enclosure. More info at controlglobal.com. **Burkert; 800/325-1405; www.burkert-usa.com**



UNCHAIN MY SENSOR

RP resistive-chain level measurement sensors are shock resistant, have an anti-stick float, and are suitable for bulk-liquid product applications. The sensors operate with a float that travels up and down a vertical stem. A permanent magnet inside the float acts on a series of closely set reed switches and resistors inside the stem. The sensor converts the resistance value into voltage, providing current signals proportional to the level of the liquid. **Scientific Technologies; 888/349-7098; www.stiapg.com.**



MAGNETOSTRICTIVE SENSOR

BW Controls 7230 explosion-proof sensor measures total level, interface level, and up to five temperatures. The



temperature sensors are distributed evenly over the active length of a stainless steel tube. It has a Modbus RTU digital output and optional analog converter, and is suitable for applications where the dielectric constant or density of the liquid is unstable. **Ametek Automation & Process Technologies; 800/635.0289; www. ametekapt.com**

PUSHBUTTON, PLUG-IN PROGRAMMING

SmartPress Model LP-D pressure/liquid level transmitters come with a 10-year warranty. The two-wire, loop



powered transmitters provide pushbutton programming, and can be supplied pre-calibrated by the factory at no added cost. An optional plug-in programming/display module, which can be moved from unit to unit, provides full setup and display capability. More info at controlglobal.com. **Princo Instruments; 800/221-9237; www.princolevelcontrols.com**

TIP, REMOTE POINT-LEVEL CONTROL

RF Series point-level indicator has no moving parts, and isn't affected by material coating. One-step calibration eliminates many steps formerly needed to calibrate and test level controls. "Test-In-Place" allows users to test level controls without removing the cover or from a remote location. This is especially important for hazardous applications or where fugitive emission releases must be prevented. **Bindicator; 800/778-9242; www.bindicator.com**

ULTRASONIC POINT-LEVEL SENSOR

Levelprox non-invasive, ultrasonic sensor provides point-level detection of liquids through metal containers, making it suitable for high pressure, hazardous, or hygienic applications. It compares the analyzed signal to empty and full conditions previously programmed.



Models T50, with a 316L stainless-steel housing and sanitary connection, and M30, with a standard 30-mm barrel, are both FM-approved for Class I, Division 2 applications. **Turck; 800/544-7769; www.turck.com**

MULTIPLE-TANK LEVEL SYSTEM

King-Gage LP3 system works in processing environments with multiple tanks or process vessels requiring level measurement and inventory monitoring. It handles up to 32 analog chan-



nels and optional RTD inputs. Transmitter inputs are scaled for specific tank geometries to generate volumetric or mass measurement. More info at controlglobal.com. **King Engineering; 800/242-8871; www.King-Gage.com**

CONVENIENT CALIBRATION

Model 5318B microprocessor-based level instrument has pushbutton cali-



bration and a digital display. For point-level operation, calibration may be accomplished without raising or lowering the process. It has an adjustable differential mode for pump on/pump off operation, is available for AC or DC power, and has a NEMA 4 explosion-proof enclosure. **Robertshaw Industrial Products; 865/981-3100; www.robertshawindustrial.com**

SELF-TESTING, ULTRASONIC SWITCHES

Echotel Model 961 single-point and 962 dual-point ultrasonic level switches require no density adjustments, work in aerated



or foamy liquids, perform self-testing of electronics, piezoelectric crystals and transducers, and check for the presence of EMI/RFI electrical noise. The switches are suitable for Safety Integrity Level (SIL) 2 loops. Options include a selection of transducer materials for various process media. **Magnetrol; 800/624-8765; www. magnetrol.com**

GOOD VIBRATIONS

Models 2110 and 2120 level switches are based on a vibrating short-fork technology that's suitable for virtually all liq-



uid applications. The switches are unaffected by flow, turbulence, bubbles, foam, vibration, or product variations. A "Fast Drip" fork means any remaining liquid is drawn away from the fork tips, allowing for quicker, more reliable detection, especially in high density liquids. More info at controlglobal.com. **Rosemount; 800/999-9307; www.rosemount.com**

WIRELESS SENSORS GO 1,000 FEET

RF10 RF event transmitters are suitable for monitoring fluid levels in tanks and drums. The portable, battery-operated units transmit up to 1,000 feet from the receiver, depending on environment. Transmitters are available in 1, 4 and 8-channel models. **APT Instruments; 877/324-5444; www.aptinstruments.com**

HOW DEEP IS YOUR SLUDGE?

Model 602 sludge-depth meter can be used in clarifiers, tanks, inclined

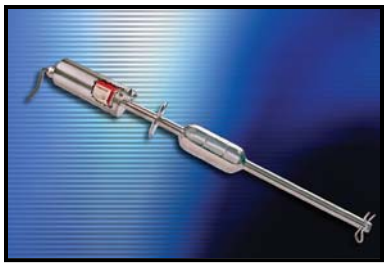


plate clarifiers (Lamellas), SBRs and dissolved-air flotation tanks. The company offers free testing of your materials at its plant, so you can prove the meter will work. **Markland Specialty Engineering; 416/244-4980; www.sludgecontrols.com**

HOT LEVEL SWITCH USES RTDS

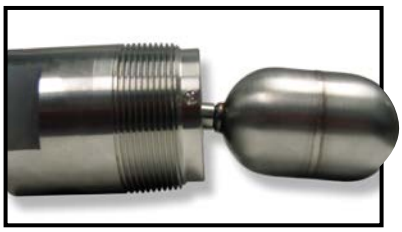
Innova-Switch level switch uses a high-resolution thermal differential technique, involving a pair of matched RTDs. The thermal differential created



between the reference and the heated RTD pair is a function of the density of the media that the sensor is contacting. It switches on a level change of 0.03 in. More info at controlglobal.com. **Sierra Instruments; 800/866-0200; www.sierrainstruments.com**

PNEUMATIC LEVEL SWITCH

Model 1540 non-bleed, side-mount pneumatic level switch is suitable for plant and OEM applications where



pneumatic valve operation is required to signal presence or absence of liquid at a discrete level. The float-operated level switch has all-316SS body and wetted parts, and a magnetic coupling. **SOR; 913/888-8150, x3303; www.sorinc.com**

MULTIVARIABLE TRANSMITTER

IMV31 provides tank-level measurement when fluid density varies. It works in open (vented) and closed (pressurized) tanks, and compensates for density changes in the tank liquid and vapor. This makes the transmitter suited for applications such as boiler drum level and others where fluid densities are affected by variations in pressure and temperature. More info at controlglobal.com. **Foxboro Measurements and Instruments; 866/746-6477; www.foxboro.com/instrumentation**

SOLID-SIGNAL RADAR SENSOR

PULS 68 two-wire radar level instrument uses a frequency of 26 GHz to achieve a strong return signal from solid materials. It's able to see and evaluate echoes in low-dielectric conditions,



and at up to 230 ft. Its swivel antenna allows for mounting to match the angle of repose created by fill or empty conditions. The sensor is unaffected by dust or buildup, angle of repose, and temperature. Applications include flour, sugar, cement, grain, plastic pellets, and aggregates. **Ohmart/Vega; 513/272-0524, x143; www.ohmartvega.com**

LONG-DISTANCE MONITOR

Sitrans LU 02 ultrasonic sensor scans liquids, solids, or a combination of both in one or two vessels of different size, shape, and configuration up to 200 ft high. It measures level, space, distance, volume or average/differential. Transducers can be mounted up to 1,200 ft from the monitor. Readings are displayed in linear engineering units on the backlit LCD. More info at controlglobal.com. **Siemens Energy & Automation; 215/646-7400 x2592; www.siemens.com**

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Smart Design Makes a Smarter Transmitter

Installation, Maintenance and Specification Made Easier by New DP Transmitter

Differential pressure flowmeters have an inherent problem—the differential pressure they produce is proportional to the square root of the flow through the primary element, and the delta-P varies over a wide range to achieve a reasonable turndown. Compounding the problem are the process pressure and ambient temperature effects on the transmitter.

A generation of maintenance techs has been driven nuts by the physical design of most differential pressure transmitters, including seals that fail, materials that corrode, circuit boards recessed deeply into small housings where fingers can't reach or where fingers and tools may all too easily contact high voltages, and case designs with displays necessarily mounted where it's difficult to see. Complex product and spare parts matrices have turned older pressure transmitter designs into real problems for procurement, stores, installation, and maintenance personnel alike.

Recent designs have addressed performance issues by improving the reference accuracy, process pressure effect, and ambient temperature coefficient of the transmitter. The ABB 364 transmitter goes considerably further by also addressing corrosion, leakage, maintenance, and procurement issues.

This unit is supplied with standard 316L stainless wetted parts and Hastelloy diaphragms to provide broad-spectrum corrosion resistance. Its stainless-steel housing protects it from the corrosive effects of its ambient environment as well. A significant feature of the housing is that the display is mounted directly on top of the housing, and can be oriented in any direction for easy observation. The terminals and circuit board aren't recessed, but are mounted at an easy-to-reach location at the top of the housing.

ABB 364's all-welded design eliminates the housing seals completely—thereby eliminating these high-potential leak paths. Eliminating the traditional O-ring seals and standardizing on stainless and Hastelloy make the 364 much easier to specify. The model's parts matrix is half as complex as its predecessors and many of its competitors, resulting in fewer spare parts requirements.

"The all-welded design of ABB 364 has allowed standardization that makes buying and installing these transmitters easier and less prone to error," says Pat Cashwell, ABB's vice president of field instrumentation. "It provides increased performance and it reduces commissioning costs with an intuitive menu structure and space-saving process connection orientation." The product will initially be available with HART followed by Profibus and Foundation fieldbus. Communication protocols can be retrofitted in the field.

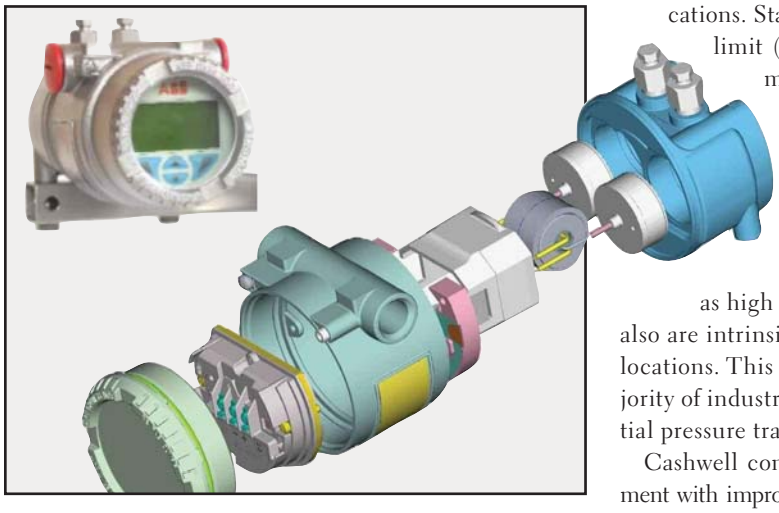
Cashwell adds, "The reference accuracy of the 364 is 0.06% of calibrated span. However testing indicates its performance is significantly better than published specifications. Stability is specified at 0.15% of upper range limit (URL) for 10 years, so users can reduce maintenance by extending time between periodic calibrations." ABB feels performance specifications will improve as production techniques are refined.

Depending on the selected range, the span of 364 models can be as low as 0.16 kPa (0.65 inches of water column) or as high as 160 bar (2,320 psi). These transmitters also are intrinsically safe, and are suitable for hazardous locations. This allows the 364 to be applicable to the majority of industrial applications for pressure and differential pressure transmitters.

Cashwell concludes, "The 364 provides a quality instrument with improved performance at a price that reduces the cost of procurement and ownership." **ABB: 215/674.6000: www.abb.com/instrumentation**

FIGURE 1.

BETTER DESIGN FROM THE BEGINNING



Radically redesigned ABB 364 differential pressure transmitter provides design, procurement and MRO ease.

Power, Usability Refine Plant Performance

Expertune's flagship product, PlantTriage, is a way to visualize the performance of a process plant. The product provides multiple views of the data that the plant control system is collecting, and permits operators and engineers to pinpoint problem areas and correct bottlenecks and maintenance issues before they cause upsets or shutdowns.

Why do you need tools like these? It used to be that all we had to do was install a control system in a plant to improve productivity, but now we need to optimize the performance of the process by tuning the control system for the highest possible profitability. Powerful software tools like PlantTriage give operators, engineers and managers the insight into their process and control systems, so they can properly tune their processes. Sometimes, however, the most powerful software tools aren't the easiest tools to use.

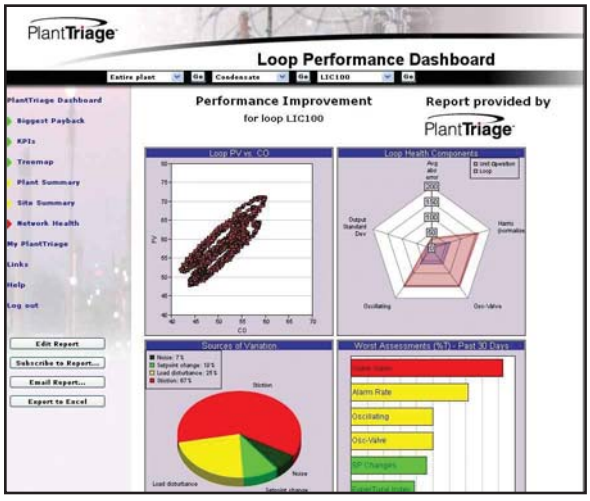
New Features

PlantTriage 6.0 is a major step forward in usability as well as power. You can begin by using existing historical data to jumpstart the project, and you can learn the system by investigating previous incidents from your own plant, using PlantTriage's full toolset.

You can configure your reports with filter and sort capability. The software provides context-sensitive drill-down for data mining, allowing you to get directly to the root cause of problems or bottlenecks, and the new Zoom capability in PlantTriage 6.0 lets the user go from the corporate data level all the way down to individual loops.

FIGURE 1.

KPIS ON THE DASH



PlantTriage 6.0's Loop Performance Dashboard shows all aspects of loop performance in one, customized page.

A Loop Performance Dashboard gives you customized, one-page detail of all aspects of loop performance. You can get a summary of overall plant performance with just a glance at the Plant Performance Dashboard. Key performance indicators (KPIs) are prominently displayed, which helps you keep your plant focused on the most important measures. This new version has Six Sigma support, including Pp and Ppk.

Solvers and Assessments

Problem-Solvers are new, pre-configured analysis tools for the most common performance problems, such as process constraints, valve failures, and loops needing tuning. Problem-Solvers for maintenance, operations, and engineering are included in the package. Some pre-configured assessment tools also are provided, including assessments of operational results, like unit cost, throughput, reliability, quality and energy costs. An Opportunity Gap assessment tool is also pre-configured, and tells the operator where to make setpoint adjustments to improve profits. According to George Buckbee of Expertune, "This is one of the fastest, easiest ways to increase profitability, similar to shifting moisture and basis weight targets on a paper machine, for example." Other operational assessments that come pre-configured with PlantTriage 6.0 include setpoint changes, mode changes, and output changes, among others. You may also define your own assessments of performance using any math you want. You can even use existing PlantTriage assessment results to calculate new user-defined assessments. You might want to define an assessment for positioner feedback, alarm information, or loop-by-loop material costs. You might want to assess operator tracking, or diagnostic information from HART, Profibus or Foundation fieldbus instruments.

Trending and Tracking

Reports can be viewed on the web, shared with co-workers, or delivered routinely via e-mail. New user-customizable report formats are supplied with PlantTriage 6.0. These include table views, trend views, and multi-trend views. The new software now integrates event information with web-based trending, so you can look at a trend, and see the operational, process, or tuning events that happened in the same time frame. Buckbee notes, "This can help you to make sense of the data, develop better process understanding, and get to root cause faster." PlantTriage 6.0 also includes some new, enhanced trending tools. You now can rescale trend on both x and y axis, scroll backward and forward in time, zoom in on a time window, and get a readout of the actual values at the hairline cursor. **Expertune Inc.; 262-369-7711; www.expertune.com/PlantTriage.html**



There are other important numbers besides a SIL rating.

The fact is, SIL ratings on their own fall short of process functional safety. That's because SIL ratings only measure the "fail safe" capability of a system. Fail Safe alone can have unintended consequences and higher risk factors associated with unplanned shutdowns. In the process industry, there are other safety-critical numbers that equate to uptime and optimal asset availability. There is only one vendor that can put up big numbers in uptime, Triconex, the process safety leader.



Only Triconex Triple Modular Redundancy technology can truly deliver the lowest lifecyle cost—by preventing unplanned shutdowns 99.999% of the time. Avoid high costs and high risk. Don't entrust the safety of your plant to lightweights whose sole idea of performance is a SIL number. Rely on Triconex, the experienced safety leader who delivers safety, uptime and the bottom line. Visit us online at triconex.com/sweepstakes by August 31, 2006, and you could win a FREE Sony® Home Theater System.

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Retirement Unleashed and Myths Continued



Greg McMillan and Stan Weiner, PE bring their wits and more than 66 years of process control experience to bear on your questions, comments, and problems. Write to them at controltalk@putman.net.

Greg: Since we have better things to do, like hang out at the pool, we're just going to expose more control and automation myths by offering some from Appendix D of *Models Unleashed*. We'll move from this serious stuff to more Groucho Marx quotes, another top 10 list, and a proposal for a new book. (Myths 1-15 appeared in "Control Mythology," Control Talk, *Control*, April '06, p. 103)

16) **You need an advanced degree to do advanced control.** Not so anymore. New software packages that form a virtual plant can automate much of the expertise needed, and eliminate the need for special interfaces. Users now can focus mostly on the application and the goal.

17) **Process simulations and model-based control are only applicable to continuous processes.** Since most of the applications are in the continuous industry, this is a common misconception. While it's true that steady-state simulations aren't valid for batch operations, since there is, by definition, no steady state, dynamic simulations can follow a batch as long as the software can handle zero flows and empty vessels. Model-based control (MPC), which looks at trajectories, is suitable for optimizing fed-batch processes. The opportunities to improve a process's efficiency by using models add up to about 25% for batch compared to 5% for continuous operations.

18) **You need consultants to maintain dynamic simulations and model-based control systems.** No longer true. The ease of use of new software allows users to get much more involved, which is critical to making sure the plant gets the most value out of the models. Previously, the benefits started to drop as soon as the consultant left the job site. Now the user should be able to tune, troubleshoot, and update the models.

19) **You don't need good operator displays and training for well-designed, advanced control systems.** Operators are the biggest constraint in most plants. Even if the models used for real-time optimization (RTO) and MPC are perfect, operators will take these systems offline if they don't understand them. The first time there's an operational problem and there's no one around to answer questions, the RTO and MPC systems are turned off—even if they're doing the right thing. Training sessions and displays should show the individual contribution of the trajectories of each controlled, disturbance, and constraint variable to the observed changes in the manipulated and optimization variables.

20) **Simple step (bump) tests are never enough. You must do a PRBS test.** A complete PRBS test may take too long. The plant may have moved to an entirely different state, tripped, or, in the case of a batch operation, finished before a PRBS test is complete. At minimum, there should be one step in each direction held to steady state. The old rule is true: if you can see



the model from a trend, then it's there. Sometimes, the brain can estimate the process gain, time delay, and lag better than a software package.

21) **You need to know your process before you start a RTO or MPC application.** This would be nice, but often a model's benefits stem from the knowledge discovered during systematic building and identification procedures. Frequently, understanding gained from developing models leads to immediate benefits such as better setpoints and instruments. Commissioning the RTO and MPC is the icing on the cake, and locks in benefits for varying plant conditions.

22) **Optimization by pushing constraints will decrease on-stream time.** Not true. MPC and RTO recognize future violations of constraints to increase on-stream time.

Groucho: These are my principles, and, if you don't like them, well…I have others.

Top 10 Signs You're Headed for Retirement

10) It takes more time to analyze your portfolio than your plant
9) Wasting time in meetings seems like good practice for wasting time in retirement.
8) Your grandkids need a control figure
7) You need more time to read and send jokes by e-mail
6) You need to spend more time with your kids to refresh your computer skills
5) The only exercise you get is walking to the coffee machine
4) Your manager is half your age.
3) Your manager asks "what's a control valve?"
2) You buy a boat
1) You need to save the Everglades more than you need to save a "flow loop"

Stan: Last year, the tropical atmosphere at Stan's place, enhanced by margaritas, led to the book *The Life and Times of an Automation Engineer.*

Groucho: From the moment I picked up our book until I laid it down, I was convulsed with laughter. Someday, I intend to read it.

Greg: This year we're drooling in anticipation of an inspiration that will lead to a book tentatively titled *The Art of Retirement—Greg and Stan's Big Book on Life in the Slow Lane.* Of course, it will have really big type and cartoons from our cohort Ted Williams. It will come with numerous heath warnings and a bib. Some chapter titles:

1) Claymation Version of Senior Living
2) Seniors on Ice
3) Senior American Idol
4) Pubs in Nursing Homes
5) Who Moved My Steak?
6) "The Present" Because I Can't Remember "The Past"
7) Single-Word Wonders—The Charo Goochie-Goochie Case History
8) Two Wild and Crazy Guys
9) Mod Squad Redux
10) Woodstock 2.5

Stan: We conclude with the Top Five Reasons to Buy the *Life and Times of an Automation Professional* and another puzzler from Hunter Vegas:

5) Charming anniversary present. This book is a diamond in the rough and says "forever" if dipped in acrylic.

4) Boss-friendly unlike previous books by Stan and Greg. There are very few jokes at the expense of management. In retrospect, this was an oversight on our part. Actually, Stan tried to add more management jokes, but Greg seems to be mellowing in anticipation of receiving Medicare.

3) Get a refund. If this book is no laughing matter, send the original UPC along with a self-addressed stamped envelope to Stan and Greg. They will email a picture of Stan cleaning the pool.

2) Discounts for futuristic "Great Automator" and "Robbie the Robot" action figures. Just show your copy at the next ISA Expo to your friendly ISA bookstore person.

1) Great preparation for the upcoming "Automation Reality" TV program. Twenty vendors and one "user" are locked in a room with 1,000 PowerPoint slides—very similar to an ISA meeting.

This Month's Puzzler: Ping, Pong, Ping, Pong, Plonk!

Twenty-four ultrasonic level transmitters were installed in a tank farm over the strong objections of the instrument contractor. After start up, some meters didn't read at all, and nearly all drifted up and down over a 24-hour period. What happened?

Send an e-mail with your answer to the Puzzler, *CONTROL* at controltalk@putman.net. Questions and comments are also welcome.

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Invensys Ups Enterprise Integration Ante

With its announcement last month of the InFusion "enterprise control system," Invensys Process Systems (IPS) formalized its bid to finally fill industry's longstanding need for a unified platform on which to truly integrate real-time controls with enterprise information systems. (Read "Invensys Launches 'First Enterprise Control System' " in this issue's news section, p. 24.)

Sound familiar? Well, it should. The editors of this magazine have spilled plenty of ink in the past on sensor-to-boardroom scenarios, starting with yours truly's first feature article profiling leading implementers of computer-integrated manufacturing (CIM). Remember that acronym? It fell out of favor when we all realized just how hard it was to make it happen! Point-to-point integration of disparate systems proved expensive, time-consuming and fragile. And too often data was simply dumped into massive archives, and did little to improve our ability to align plant performance with business priorities.



It's too early to tell the Visicalcs of today from the Microsoft Offices of tomorrow, but, for now, Invensys Process Systems appears headed in a positive direction.

Beyond that, this kind of data is of really little value unless it is used, and management level business leaders are just now beginning to understand the value of the types of data that CIM and its successors can provide.

"Most of those ideas never really got past Powerpoint," confessed controls industry veteran Rick Bullotta, now of SAP Labs, on hand at InFusion's launch to lend the enterprise software giant's support to the initiative. "However, I think we're finally there."

So, just what has changed since the CIM days that might make this round anything but more marketing-speak? First, the key standards and technology enablers are finally mature enough to make integration relatively less painful and more robust. For example, our industry's virtual standardization on SAP at the enterprise layer and Microsoft for IT infrastructure already has greatly simplified the task at hand.

In the case of InFusion, Microsoft's .NET and BizTalk Server as well as SAP's NetWeaver and xMII technologies are leveraged to streamline integration and visualization tasks. Consensus standards such as Open Operations & Maintenance (O&M)—the convergence of the OPC, ISA 95 and MIMOSA standards—also play a key role in reducing integration effort "by at least 50%," says IPS president Mike Caliel.

At the plant-floor level, one way to think of InFusion is as an extension of the robustness of the Foxboro I/A distributed control system architecture to include legacy devices that once would have required point-to-point integration. The clever thing that Invensys has done is to take the industry-leading cadre of device drivers developed for its Wonderware InTouch platform, and then add a layer of object abstraction on top through which to implement more advanced performance management applications. In effect, non-Invensys systems can now be treated as objects within the distributed "Enterprise Control System" environment. Not rocket science in concept; they're simply making integration much easier to do and maintain than it used to be.

Another reason for more widespread integration of controls and enterprise systems is an eminently practical one, having to do with market dynamics. Control itself has been for the most part commoditized, and the next battle for differentiation among the major automation suppliers will have to be engaged at the execution and integration level. At this level, token "openness," which has long been the motherhood-and-apple-pie value of the automation community, will give way to vendor standardization at the integration and execution level, further improving the effectiveness of these efforts.

Just look at the enterprise software space. A scant few years ago, a multitude of best-of-breed application providers vied for dominance in narrow specializations. As the market matured, however, the advantages of best-of-breed gave way to application suites with built-in, cross-application integration.

Similarly, all the major global automation players will continue to develop their own integrated suites of tools designed to optimize overall manufacturing asset performance. It's too early to tell the Visicalcs of today from the Microsoft Offices of tomorrow, but, for now, Invensys Process Systems appears headed in a positive direction.

Keith Larson,
VP Content, Putman Media

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